2 Geometry and Measurement

The big idea. Geometry is the study of *shapes*; measurement attaches *numbers* to those shapes: lengths, angle measures, perimeters, areas and volumes. Almost every problem in this topic is solved with a handful of very simple ideas that are used over and over again:

- **Angles have a fixed "budget":** two supplementary angles always add up to 180°, the three angles of a triangle always add up to 180°, and a full turn is 360°. Knowing some of the angles, we can work out the rest.
- **Areas can be cut, rearranged and subtracted:** a complicated shape is split into familiar pieces (rectangles, triangles, trapezoids), or computed as "big shape minus the extra bits".
- **Ratios instead of numbers:** "what percentage is shaded" only requires comparing the shaded part with the whole shape – often without knowing any real sizes.
- **Shape does not change when turned or flipped:** rotating, reflecting or rolling a cube only changes *position*, never *size* or *shape*.
- **A box is built in layers:** volume = base area × height.

Whenever you meet a geometry problem, always **draw the figure and write the given numbers on it**, then ask yourself: "What do I know? What is missing? Which formula connects the two?".

① The coordinate plane

The coordinate system Oxy consists of two number lines that are perpendicular at the origin O:

- the horizontal axis Ox is called the **x-axis**; its positive direction points to the *right*;
- the vertical axis Oy is called the **y-axis**; its positive direction points *up*.

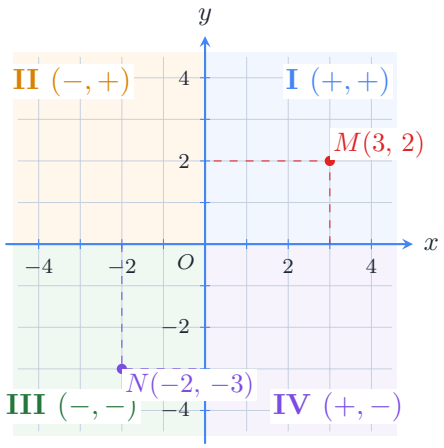
Each point M of the plane corresponds to exactly one pair of numbers (x_0, y_0), written $M(x_0, y_0)$:

- x_0 is the **x-coordinate**: drop a perpendicular from M to the x-axis; its foot is at the number x_0;
- y_0 is the **y-coordinate**: draw a perpendicular from M to the y-axis; its foot is at the number y_0.

The two axes divide the plane into four **quadrants**, numbered counterclockwise:

	Quadrant I	Quadrant II	Quadrant III	Quadrant IV
Signs of (x, y)	$(+, +)$	$(-, +)$	$(-, -)$	$(+, -)$

Distances along horizontal and vertical lines. Two points with the same y-coordinate (on the same horizontal line) are a distance apart equal to the difference of their x-coordinates (larger minus smaller). Similarly, two points with the same x-coordinate are a distance apart equal to the difference of their y-coordinates.



ThinkAlgo Tip. Read coordinates in the order "**across first, then up or down**": start at O, move right (positive) or left (negative) to get the x-coordinate; then move up (positive) or down (negative) to get the y-coordinate. For example, to reach $N(-2, -3)$: go 2 squares left, then 3 squares down.

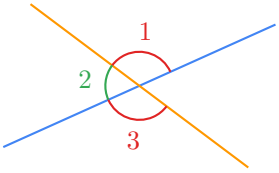
Common mistake. Swapping the two numbers is a very common mistake: $(3, -1)$ and $(-1, 3)$ are two *different* points (one in quadrant IV, the other in quadrant II). The x-coordinate is *always* written first.

Mini example. Let $A(1, 1)$, $B(5, 1)$, $C(5, 4)$. Since A and B have the same y-coordinate 1, $AB = 5 - 1 = 4$. Since B and C have the same x-coordinate 5, $BC = 4 - 1 = 3$. Angle B is a right angle (one side horizontal, one side vertical), so the area of triangle ABC is $\dfrac{4 \cdot 3}{2} = 6$ (square units).

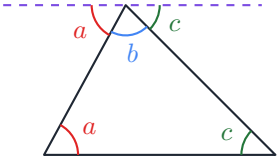
Note for parents. You can turn the coordinate plane into a "treasure hunt" on squared paper: you say "left 2, up 3", your child marks the point and writes its coordinates $(-2, 3)$. Then swap roles: your child gives directions and you draw. After a few rounds, the order "x first, y second" becomes automatic.

② Angles: supplementary, vertically opposite, angle sum of a triangle

- A **right angle** is 90°, a **straight angle** (half a turn) is 180°, a **full turn** is 360°.
- **Two adjacent supplementary angles** (sharing one arm, with the other two arms forming a straight line) add up to 180°.
- **Two vertically opposite angles** (each arm of one is the extension of an arm of the other) are **equal**.
- **The three angles of a triangle add up to** 180°**.** Hence: third angle = 180° − (sum of the two known angles).
- An **exterior angle of a triangle** (supplementary to an interior angle) equals **the sum of the two interior angles not adjacent to it**.
- In a right triangle, the two acute angles add up to 90°.



$\widehat{1}+\widehat{2}=180°$ (supplementary)
$\widehat{1}=\widehat{3}$ (vertically opposite)



$a+b+c=180°$

Why is it true? **Vertically opposite angles are equal.** In the left figure, $\widehat{1}+\widehat{2}=180°$ (supplementary) and $\widehat{3}+\widehat{2}=180°$ (also supplementary). Angles $\widehat{1}$ and $\widehat{3}$ are both supplementary to the same angle $\widehat{2}$, so $\widehat{1}=180°-\widehat{2}=\widehat{3}$.

The angles of a triangle add up to $180°$**.** Through the top vertex of the triangle, draw the line parallel to the base (dashed). For parallel lines, *alternate angles* are equal: the angle on the left at the top vertex equals a, and the angle on the right equals c. The three angles a, b, c at the top vertex together form a straight angle, so $a+b+c=180°$.

Exterior angle. The exterior angle at a vertex is supplementary to the interior angle there: exterior angle $=180°-$ interior angle. But $180°-$ interior angle is exactly the sum of the other two interior angles. So the exterior angle equals the sum of the two interior angles not adjacent to it.□

ThinkAlgo Tip. In "find the angle" problems, write the measure of every angle you find **on the figure** immediately. An angle you have just found is often the key to the next one (supplementary, vertically opposite, angle sum, ...).

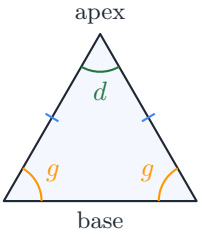
Common mistake. Do not confuse *vertically opposite* angles (equal) with *supplementary* angles (sum $180°$). Vertically opposite angles sit "across from each other" at the intersection point and share no arm; adjacent supplementary angles sit "next to each other" and share an arm.

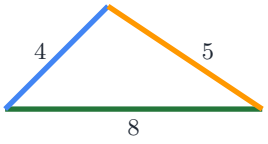
③ Isosceles triangles, equilateral triangles and the triangle inequality

- An **isosceles triangle** has two equal sides (the *legs*); the third side is the *base*, and the angle between the two legs is the *apex angle*.
- In an isosceles triangle, **the two base angles are equal**. So if the apex angle is d and each base angle is g, then

$$d+2g=180°, \qquad d=180°-2g, \qquad g=\frac{180°-d}{2}.$$

- A base angle is always **acute** (less than $90°$), because the two base angles are equal and their sum must be less than $180°$.
- An **equilateral triangle** has three equal sides and three equal angles, each $180°\div 3=60°$.
- **Triangle inequality:** in a triangle, each side is always *less* than the sum of the other two. To check whether three lengths form a triangle, it is enough to check: *longest side* $<$ *sum of the other two.*





$8 < 4 + 5$: a triangle exists

Why is it true? **Why are the base angles equal?** Let M be the midpoint of the base BC of the isosceles triangle ABC ($AB = AC$). Triangles ABM and ACM have $AB = AC$, $BM = CM$ and the common side AM, so they are congruent (side – side – side). Hence the corresponding angles $\widehat{B}$ and $\widehat{C}$ are equal.

Why does the triangle inequality hold? The shortest path between two points is the straight segment. Going from B to C "via" A (that is, travelling $BA + AC$) is always longer than going straight along BC. If the longest side were equal to or greater than the sum of the other two, the two short sides could not "reach each other": they would lie flat along the long side or fail to meet, and no triangle would be formed. □

Common mistake. When a problem only says "an isosceles triangle has *an angle* of 50°", we do **not yet know** whether it is the apex angle or a base angle. We must consider **both cases**:

- 50° is the apex angle: the base angles are $(180° - 50°) \div 2 = 65°$ and 65°;
- 50° is a base angle: the other base angle is also 50°, and the apex angle is $180° - 2 \cdot 50° = 80°$.

Only when the given angle is *greater than or equal to* 90° must it be the apex angle.

④ Rotations, reflections and rolling cubes

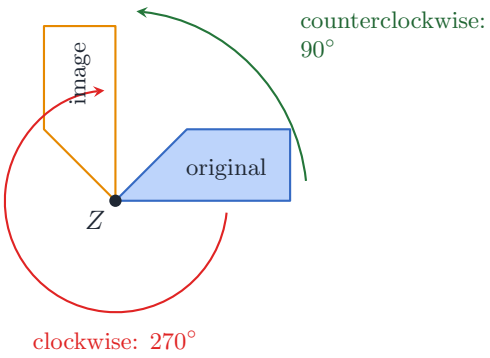
A **rotation** about a point Z (the *centre of rotation*) through an angle α in a given direction (clockwise or counterclockwise) sends each point M to a point M' such that $ZM' = ZM$ and $\widehat{MZM'} = \alpha$.

- Every point of the figure turns through **the same angle**, in the same direction; distances to the centre do not change.
- A rotation keeps the shape and size of the figure; the figure is only "turned", never "flipped over".
- Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise. A 360° rotation brings the figure back to where it started.
- **How to measure the angle of rotation:** choose a segment starting at the centre Z on the original figure, find the matching segment on the new figure, and measure the angle that segment has swept through *in the given direction*.

A **reflection in a line** (flipping over the line, like a mirror) also keeps sizes, but **reverses the orientation** of the figure: the letter P seen in a mirror becomes a backwards P.

In the coordinate plane (centre of rotation at the origin O):

Rotation 90° counterclockwise	$(x, y) \to (-y, x)$
Rotation 180°	$(x, y) \to (-x, -y)$
Rotation 90° clockwise	$(x, y) \to (y, -x)$
Reflection in the x-axis / in the y-axis	$(x, y) \to (x, -y)$ / $(x, y) \to (-x, y)$



Why is it true? Why is a 270° clockwise rotation the same as a 90° counterclockwise one? A full turn is 360°. After going 270° one way, only $360° - 270° = 90°$ more would bring us back to the start; so the end position is exactly where we would arrive by going 90° the other way. In the figure: to turn the horizontal side (pointing right) into the vertical side (pointing up), we can rotate 90° counterclockwise, or rotate 270° clockwise (through pointing down, pointing left, and only then up). □

Rolling a cube. When a cube is rolled, each face is carried along unchanged. So a face with a painted pattern:

- may appear in another position (front, top, side, ...);
- may be *turned* by 90°, 180° or 270° within its own plane;
- **but never** becomes a mirror image, changes size, or "gains" extra paint.

ThinkAlgo Tip. For cube problems, "mark" one special feature of the pattern (for example: the right angle of the black triangle sits at *a corner* of the face) and check that feature in every option. Cutting out a paper square, drawing the pattern on it and turning it is also a very effective check.

⑤ Units of measurement and the number line

Units of length. Each unit is 10 times the next one:

$$1 \text{ km} = 10 \text{ hm}, \quad 1 \text{ hm} = 10 \text{ dam}, \quad 1 \text{ dam} = 10 \text{ m},$$

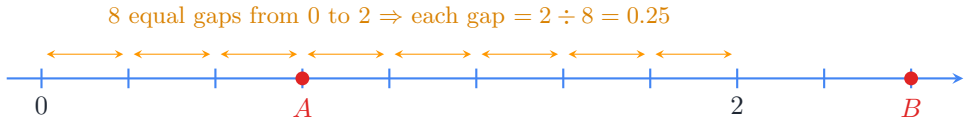
$$1 \text{ m} = 10 \text{ dm}, \quad 1 \text{ dm} = 10 \text{ cm}, \quad 1 \text{ cm} = 10 \text{ mm}.$$

The most used relations: $1 \text{ m} = 100 \text{ cm} = 1000 \text{ mm}$; $1 \text{ km} = 1000 \text{ m}$.

Units of area go up by a factor of 100: $1\ \text{m}^2 = 100\ \text{dm}^2 = 10\,000\ \text{cm}^2$.
Units of volume go up by a factor of 1000: $1\ \text{m}^3 = 1000\ \text{dm}^3$, $1\ \text{dm}^3 = 1000\ \text{cm}^3 = 1$ litre.
Conversion rule: from a *larger* unit to a *smaller* unit, **multiply**; from a *smaller* unit to a *larger* unit, **divide**.
Number line with equally spaced ticks. If we know the numbers at two ticks and count k *gaps* (not k ticks) between them, then

$$\text{value of one gap} = \frac{\text{difference of the two numbers}}{k}.$$

Why is it true? Why is $1\ \text{m}^2 = 10\,000\ \text{cm}^2$ and not $100\ \text{cm}^2$? One square metre is the area of a square with side 1 m, i.e. side 100 cm. That square can be divided into 100 rows of 100 squares of side 1 cm, so it contains $100 \times 100 = 10\,000$ squares of $1\ \text{cm}^2$. Likewise, a cube of side 1 dm = 10 cm consists of $10 \times 10 \times 10 = 1000$ small cubes of $1\ \text{cm}^3$. □



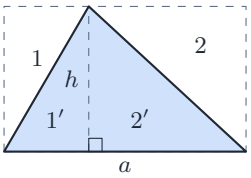
Mini example. On the number line above, A is three gaps from 0, so $A = 3 \cdot 0.25 = 0.75$; B is ten gaps from 0, so $B = 10 \cdot 0.25 = 2.5$.

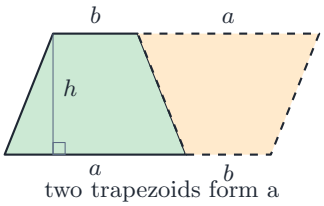
Common mistake. Counting *ticks* instead of *gaps*: from tick 0 to tick 2 in the figure above there are 9 ticks but only 8 gaps. Between two numbers there is always one gap fewer than ticks (like trees in a row: 9 trees have 8 spaces between them).

⑥ Perimeter and area of plane figures

Figure	Perimeter	Area
Rectangle (length a, width b)	$P = 2 \cdot (a + b)$	$S = a \cdot b$
Square (side a)	$P = 4 \cdot a$	$S = a \cdot a = a^2$
Triangle (base a, height h to that base)	sum of the three sides	$S = \frac{a \cdot h}{2}$
Parallelogram (base a, height h)	2·(sum of two adjacent sides)	$S = a \cdot h$
Trapezoid (parallel sides a, b; height h)	sum of the four sides	$S = \frac{(a + b) \cdot h}{2}$

The **height** of a triangle (parallelogram, trapezoid) is the segment drawn *perpendicular* from the vertex (or from the opposite side) to the line containing the base.
Working backwards: from $S = \frac{a \cdot h}{2}$ we get $h = \frac{2 \cdot S}{a}$ and $a = \frac{2 \cdot S}{h}$; from the perimeter of a square, side $= P \div 4$.





two trapezoids form a parallelogram with base $a + b$

Why is it true? **Area of a triangle.** Place the triangle inside the rectangle with the same base a and the same height h (left figure). The height splits the rectangle into two smaller rectangles; each smaller rectangle is cut by a diagonal into two equal triangles: the part inside the triangle ($1'$, $2'$) equals the part outside it (1, 2). So the triangle is exactly half of the rectangle: $S = \dfrac{a \cdot h}{2}$.
Area of a parallelogram. Cut off a right triangle at one end and slide it to the other end: we get a rectangle with the same base a and the same height h: $S = a \cdot h$.
Area of a trapezoid. Two identical trapezoids, one turned upside down, fit together into a parallelogram with base $a+b$ and height h (right figure). One trapezoid is half of it: $S = \dfrac{(a+b) \cdot h}{2}$.
□

ThinkAlgo Tip. Two triangles with **the same base** (or equal bases) and **the same height** have equal areas, even if they look very different. Consequence: the segment joining a vertex to the midpoint of the opposite side divides a triangle into two parts of equal area.

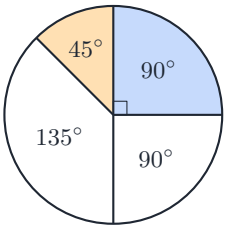
Common mistake.

- The height must be **perpendicular** to the base; a slanted side is *not* the height.
- Forgetting to divide by 2 in the area formulas of the triangle and the trapezoid.
- Mixing up units: perimeter is measured in cm, area in cm^2. A square with perimeter 36 cm has side 9 cm and area 81 cm^2 – not 36 cm^2.

⑦ Parts of an area: fractions, percentages, circles and rearranging

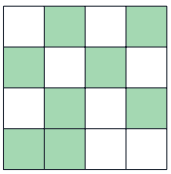
- If a figure is divided into n **equal** parts and m parts are shaded, the shaded part is $\dfrac{m}{n}$ of the figure, i.e. $\dfrac{m}{n} \cdot 100\%$.
- **Circles and sectors.** One turn around the centre is 360°. A sector with central angle $n°$ is $\dfrac{n}{360}$ of the area of the circle. For example: a semicircle corresponds to 180°, a quarter circle to 90° (a right angle at the centre).
- **Spinners:** the pointer stops at random, all directions being equally likely, so the chance of stopping in a region equals the fraction of the area (i.e. of the angle) of that region.
- **Cut and rearrange:** split the figure into familiar pieces and add their areas. **Complement:** required area = area of the big figure − area of the extra pieces.
- **Systematic counting of figures:** when asked how many figures (triangles, parallelograms, ...) appear in a diagram, sort them by *size* (made of 1 piece, 2 pieces, 3 pieces, ...) and count

each type so that nothing is missed or counted twice.



blue: $\frac{90}{360} = \frac{1}{4} = 25\%$

orange: $\frac{45}{360} = \frac{1}{8} = 12.5\%$



8 of 16 squares: $\frac{8}{16} = 50\%$

Why is it true? Why is a sector with angle n° equal to $\frac{n}{360}$ of the circle? Imagine cutting the circle into 360 tiny sectors, each with a central angle of 1°. These tiny sectors are identical (turning the circle by 1° about its centre makes one fit exactly onto the next), so they have equal areas, each $\frac{1}{360}$ of the circle. A sector with angle n° is made of n such tiny sectors. □

Common mistake. You may only count "m parts out of n" when the parts are **equal**. If the pieces have different sizes, reduce them to one kind of piece (for example by cutting them further into equal pieces) or compute the area of each piece.

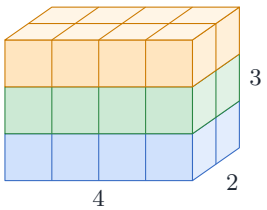
Note for parents. Cut a cake (or a paper circle) into 4, then into 8 equal parts so your child can see: 1 quarter = 90°, 1 eighth = 45°, and half of a quarter is an eighth. The idea "halving the angle halves the area" becomes very concrete.

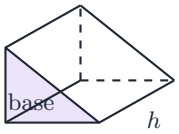
⑧ Solids: volume and surface area

- A **rectangular box** (rectangular prism) with dimensions a, b, c:

$$V = a \cdot b \cdot c, \qquad S_{\text{lateral}} = 2 \cdot (a + b) \cdot c, \qquad S_{\text{total}} = 2 \cdot (ab + bc + ca).$$

- A **cube** with side a: $V = a^3$, $S_{\text{total}} = 6 \cdot a^2$.
- A **right prism** (with a triangular, quadrilateral, ... base): $V = S_{\text{base}} \cdot h$, where h is the height of the prism (the length of a lateral edge).
- The number of 1 cm × 1 cm × 1 cm cubes needed to fill a box exactly equals the volume of the box in cm^3.





each layer $4 \cdot 2 = 8$ cubes, 3 layers: $8 \cdot 3 = 24$ cubes $V = S_{\text{base}} \cdot h$

Why is it true? **Why is** $V = a \cdot b \cdot c$**?** Stack 1 cm^3 cubes in layers. The bottom layer is a rectangle of a rows $\times$ b columns, containing $a \cdot b$ cubes (exactly the base area). Stack c such layers and we have $a \cdot b \cdot c$ cubes. The argument "each layer = base area, number of layers = height" works for every right prism, so $V = S_{\text{base}} \cdot h$. In particular, for a prism whose base is a right triangle with legs a, b: two such prisms fit together into an $a \times b \times h$ box, so each prism has volume $\dfrac{a \cdot b}{2} \cdot h$. $\square$

ThinkAlgo Tip. The surface area of a box is the total area of its 6 faces, which come in 3 pairs of equal opposite faces. When small blocks are joined into a big block, the faces that are "pressed together" are no longer on the surface, so the surface area of the big block is *less* than the total surface area of the small blocks.

Example 1

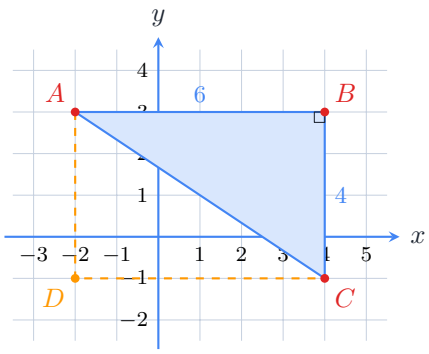
In the coordinate plane Oxy, consider the points $A(-2, 3)$, $B(4, 3)$ and $C(4, -1)$.

a) In which quadrant does each of the points A, B, C lie?

b) Find the lengths AB, BC and the area of triangle ABC.

c) Find the coordinates of the point D such that $ABCD$ is a rectangle.

Guided solution.



Step 1. Use the signs of the coordinates to find the quadrant

$A(-2, 3)$: negative x, positive y $\Rightarrow$ quadrant II. $B(4, 3)$: both positive $\Rightarrow$ quadrant I. $C(4, -1)$: positive x, negative y $\Rightarrow$ quadrant IV.

Step 2. Find the lengths of the horizontal and vertical sides

A and B have the same y-coordinate 3, so AB is horizontal and $AB = 4 - (-2) = 4 + 2 = 6$ (units). B and C have the same x-coordinate 4, so BC is vertical and $BC = 3 - (-1) = 3 + 1 = 4$.

Step 3. Find the area of the triangle

Side AB is horizontal and side BC is vertical, so angle B is a right angle. The area of a right triangle is half the product of the two legs:

$$S_{ABC} = \frac{AB \cdot BC}{2} = \frac{6 \cdot 4}{2} = 12 \text{ (square units)}.$$

Step 4. Find the point D

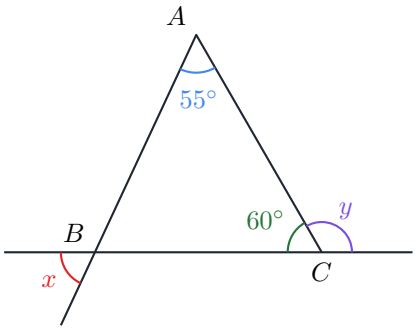
In rectangle $ABCD$, D is "opposite" B: D must be vertically in line with A (same x-coordinate -2) and horizontally in line with C (same y-coordinate -1). So $D(-2, -1)$, in quadrant III.

Check: Rectangle $ABCD$ has area $6 \cdot 4 = 24$, exactly twice the area of triangle ABC (the diagonal AC cuts the rectangle into two equal halves).

Answer: **a)** A: **II**, B: **I**, C: **IV**. **b)** $AB = 6$, $BC = 4$, $S = 12$. **c)** $D(-2, -1)$.

Example 2

In the figure below, triangle ABC has $\widehat{A} = 55°$, $\widehat{C} = 60°$. Sides AB and CB are extended beyond B; side BC is extended beyond C. Find the measures x and y.



Guided solution.

Analysis. Angle x lies *outside* the triangle, but it is *vertically opposite* to the angle $\widehat{ABC}$ of the triangle (the arms of x are the extensions of BC and of BA). So we find $\widehat{ABC}$ first.

Step 1. **Find angle B of the triangle**

Recall. The three angles of a triangle add up to 180°.

$\widehat{ABC} = 180° - (55° + 60°) = 180° - 115° = 65°$.

Step 2. **Find x**

Recall. Vertically opposite angles are equal.

Angle x and angle $\widehat{ABC}$ are vertically opposite, so $x = 65°$.

Step 3. **Find y**

Angle y is supplementary to angle $\widehat{ACB}$, so $y = 180° - 60° = 120°$.

Check: By the exterior angle property, y must equal the sum of the two non-adjacent interior angles: $55° + 65° = 120°$. It matches!

Answer: $x = 65°$, $y = 120°$.

Example 3

a) An isosceles triangle has an angle of 40°. Find the other two angles (consider all possibilities).

b) An isosceles triangle has an angle of 100°. Find the other two angles.

c) Which of the triples of lengths (3, 4, 8) and (3, 5, 7) forms a triangle? An isosceles triangle has two sides of lengths 4 cm and 9 cm. Find its perimeter.

Guided solution.

Step 1. Part a – two cases

Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°.

Case 1: 40° *is the apex angle.* The base angles are equal and add up to $180° - 40° = 140°$, so each is $140° \div 2 = 70°$. The triangle has angles 40°, 70°, 70°.
Case 2: 40° *is a base angle.* The other base angle is also 40°; the apex angle is $180° - 40° - 40° = 100°$. The triangle has angles 40°, 40°, 100°.

Step 2. Part b – only one case

100° cannot be a base angle: otherwise the two base angles alone would add up to $200° > 180°$. So 100° is the apex angle, and each base angle is $(180° - 100°) \div 2 = 40°$.

Step 3. Part c – use the triangle inequality

Recall. Three lengths form a triangle exactly when the longest is less than the sum of the other two.

(3, 4, 8): $8 > 3 + 4 = 7$, *no* triangle. (3, 5, 7): $7 < 3 + 5 = 8$, a triangle exists.
An isosceles triangle with two sides 4 and 9 has its third side equal to 4 or to 9:

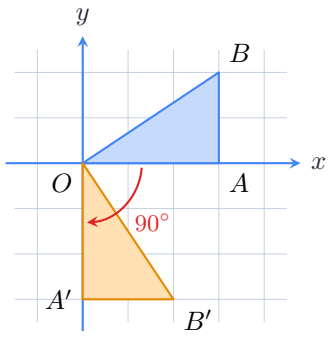
- (4, 4, 9): $9 > 4 + 4 = 8$, rejected;
- (4, 9, 9): $9 < 4 + 9 = 13$, fine. Perimeter $= 4 + 9 + 9 = 22$ cm.

Answer: a) 70° and 70°, or 40° and 100°. b) 40° and 40°. c) only (3, 5, 7); perimeter 22 cm.

Example 4

Triangle OAB with $O(0, 0)$, $A(3, 0)$, $B(3, 2)$ is rotated about the origin O *clockwise* and becomes triangle $OA'B'$ with $A'(0, -3)$, $B'(2, -3)$. What is the angle of rotation? If we rotate counterclockwise instead, through what angle must we turn to get the same result?

Guided solution.



Step 1. Choose a segment starting at the centre of rotation

Segment OA lies along the ray Ox, pointing *right*. Its image is OA' with $A'(0, -3)$, lying on the y-axis and pointing *down*.

Step 2. Measure the angle swept clockwise

A clock hand going from the "3 o'clock" direction (right) to the "6 o'clock" direction (down) sweeps

a right angle. So the clockwise angle of rotation is $90°$.

Step 3. Convert to the other direction

Recall. Rotating α clockwise gives the same result as rotating $360° - \alpha$ counterclockwise.

Counterclockwise: $360° - 90° = 270°$.

Check: By the table in section 4, a $90°$ clockwise rotation sends (x, y) to $(y, -x)$: $A(3, 0) \to (0, -3) = A'$ and $B(3, 2) \to (2, -3) = B'$. Correct.

Answer: $90°$ clockwise, equivalent to $270°$ counterclockwise.

Example 5

On a number line, the ticks are equally spaced. The first tick is at 0, the seventh tick is at 3 (that is, six gaps away from 0). Point M is at the tick three gaps from 0, point N at the tick nine gaps from 0.

a) Find the numbers at M and N, and compute $M \div N$.

b) If the unit on the line is the metre, convert M and N to centimetres.

Guided solution.



Step 1. Find the value of one gap

From 0 to 3 there are 6 equal gaps, so each gap is worth $3 \div 6 = 0.5$.

Step 2. Find M and N

M is three gaps from 0: $M = 3 \cdot 0.5 = 1.5$. N is nine gaps from 0: $N = 9 \cdot 0.5 = 4.5$.

Step 3. Compute the quotient

$M \div N = 1.5 \div 4.5 = \dfrac{15}{45} = \dfrac{1}{3} \approx 0.33$. (Quick view: N is 3 times as far from 0 as M, so $M \div N = \frac{1}{3}$.)

Step 4. Convert units

Recall. 1 m = 100 cm; from a larger unit to a smaller unit, multiply.

$1.5 \text{ m} = 1.5 \cdot 100 \text{ cm} = 150 \text{ cm}$; $4.5 \text{ m} = 450 \text{ cm}$.

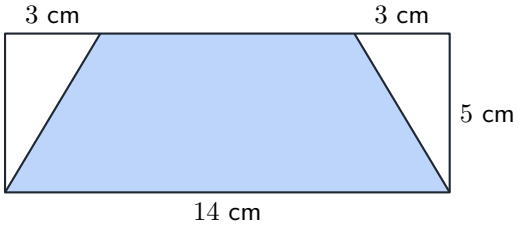
Answer: a) $M = 1.5$; $N = 4.5$; $M \div N = \frac{1}{3}$. b) 150 cm and 450 cm.

Example 6

A rectangle is 14 cm long and 5 cm wide. On its top side, two points are marked, each 3 cm from a top corner; they are joined to the two bottom corners to form the shaded trapezoid (see figure).

a) Find the area of the trapezoid in two ways.

b) A triangle has area 24 cm^2 and a base of length 8 cm. Find the height to that base.



Guided solution.

Step 1. **Method 1 – the trapezoid formula**

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$, where a, b are the parallel sides and h is the height.

Longer parallel side $a = 14$ cm. Shorter parallel side $b = 14 - 3 - 3 = 8$ cm. The height equals the width of the rectangle: $h = 5$ cm.

$$S = \frac{(14+8)\cdot 5}{2} = \frac{22\cdot 5}{2} = \frac{110}{2} = 55 \text{ (cm}^2\text{)}.$$

Step 2. **Method 2 – big shape minus the extra pieces**

Area of the rectangle: $14\cdot 5 = 70$ (cm^2). The two unshaded parts are right triangles, each with legs 3 cm and 5 cm and area $\dfrac{3\cdot 5}{2} = 7.5$ (cm^2). Area of the trapezoid: $70 - 2\cdot 7.5 = 70 - 15 = 55$ (cm^2).

Step 3. **Part b – find the height by working backwards**

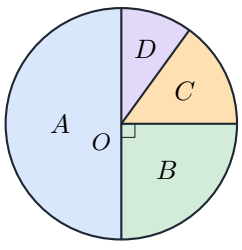
From $S = \dfrac{a\cdot h}{2}$ we get $h = \dfrac{2\cdot S}{a} = \dfrac{2\cdot 24}{8} = \dfrac{48}{8} = 6$ (cm).

Check: $\dfrac{8\cdot 6}{2} = 24$. Correct.

Answer: a) 55 cm^2. b) 6 cm.

Example 7

A circular spinner with centre O is divided into four regions A, B, C, D: region A is a semicircle, region B is a quarter circle, and the rest is split into regions C and D so that the central angle of C is one and a half times the central angle of D. What percentage of the spinner's area does each region take up? About what percentage of spins will stop in region C?



Guided solution.

Step 1. **Central angles of regions A and B**

The whole circle corresponds to 360°. Region A (a semicircle) corresponds to 180°, region B (a

quarter) to 90°.

Step 2. Central angles of regions C and D

The remainder: $360° - 180° - 90° = 90°$. We must split 90° into two parts with C one and a half times D; that is, if D is 2 equal parts then C is 3 such parts. Together 5 parts correspond to 90°, so one part is $90° \div 5 = 18°$. Hence the angle of D is $2 \cdot 18° = 36°$ and the angle of C is $3 \cdot 18° = 54°$.

Step 3. Convert to percentages

> **Recall.** A sector with central angle $n°$ is $\dfrac{n}{360}$ of the circle.

A: $\dfrac{180}{360} = 50\%$; B: $\dfrac{90}{360} = 25\%$; C: $\dfrac{54}{360} = 0.15 = 15\%$; D: $\dfrac{36}{360} = 0.1 = 10\%$.

Check: $50\% + 25\% + 15\% + 10\% = 100\%$, and $54° = 1.5 \cdot 36°$. Correct.

Answer: A: 50%, B: 25%, C: 15%, D: 10%; about 15% of spins stop in region C.

Example 8

a) How many cubes with edge 1 cm are needed to fill a 5 cm × 3 cm × 2 cm box? Find the total surface area of the box.

b) A right prism has a base that is a right triangle with legs 3 cm and 4 cm, and the prism is 10 cm high. Find its volume. If a prism with the same base has volume 90 cm^3, what is its height?

Guided solution.

Step 1. Part a – count layer by layer

> **Recall.** $V_{\text{box}} = a \cdot b \cdot c$; $S_{\text{total}} = 2 \cdot (ab + bc + ca)$.

The bottom layer has $5 \cdot 3 = 15$ cubes; there are 2 layers, so we need $15 \cdot 2 = 30$ cubes (volume 30 cm^3).

The three pairs of opposite faces have areas $5 \cdot 3 = 15$, $3 \cdot 2 = 6$, $5 \cdot 2 = 10$ (cm^2), so $S_{\text{total}} = 2 \cdot (15 + 6 + 10) = 2 \cdot 31 = 62$ (cm^2).

Step 2. Part b – base area times height

> **Recall.** $V_{\text{prism}} = S_{\text{base}} \cdot h$.

The base is a right triangle: $S_{\text{base}} = \dfrac{3 \cdot 4}{2} = 6$ (cm^2). Volume: $V = 6 \cdot 10 = 60$ (cm^3).

With volume 90 cm^3: $h = V \div S_{\text{base}} = 90 \div 6 = 15$ (cm).

Check: $6 \cdot 15 = 90$. Correct.

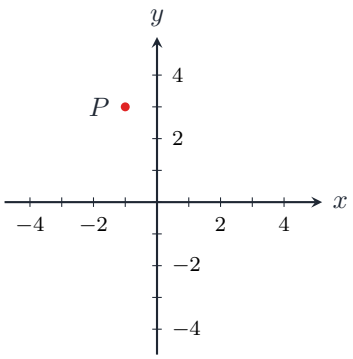
Answer: a) 30 cubes; 62 cm^2. b) 60 cm^3; 15 cm.

Note for parents. After each example, ask your child to *say in words* the formula just used and *why* it is true (for example: "a triangle is half of the rectangle with the same base and height"). Understanding the "why" means less rote memorisation and fewer mistakes on unfamiliar figures.

Core practice

Problem 1

In the coordinate plane below, each tick on the two axes represents 1 unit. Which pair of numbers could be the coordinates of point P?



(A) (1, 3) (B) (1, −3) (C) (−3, 1) (D) (3, −1) (E) (−1, 3)

Detailed solution

Understanding the problem. We read the position of P relative to the two axes to find the *sign* and the *size* of its coordinates, then compare with the five options.

> **Recall.** For a point $M(x_0, y_0)$: x_0 tells whether the point is to the right ($x_0 > 0$) or to the left ($x_0 < 0$) of the y-axis; y_0 tells whether it is above ($y_0 > 0$) or below ($y_0 < 0$) the x-axis. The x-coordinate is always written first.

Step 1. Find the signs of the coordinates

P is to the *left* of the y-axis, so its x-coordinate is **negative**. P is *above* the x-axis, so its y-coordinate is **positive**. So P is in quadrant II, of the form $(-, +)$.

Step 2. Eliminate options

- (A) (1, 3): form $(+, +)$, quadrant I – rejected.
- (B) (1, −3): form $(+, -)$, quadrant IV – rejected.
- (D) (3, −1): form $(+, -)$ – rejected.
- Remaining: (C) (−3, 1) and (E) (−1, 3), both of the form $(-, +)$.

Step 3. Compare the sizes

Drop a perpendicular from P to the x-axis: its foot lies between 0 and −2, exactly at tick −1. Draw a perpendicular to the y-axis: its foot lies between 2 and 4, exactly at tick 3. So P is very close to the y-axis (1 unit away) and fairly high up (3 units above the x-axis): $P(-1, 3)$.
The point (−3, 1) in option (C) is the opposite: far from the y-axis (3 units) and low (1 unit), which does not match the figure.

Common mistake. Option (C) is the "trap" for those who read the y-coordinate first. Remember: *across first, then up or down.*

Check: From O go 1 tick left, then 3 ticks up: we arrive exactly at P.
So the answer is **(E)**.

Answer: (E) $(-1, 3)$

Problem 2

Two angles of a triangle measure 25° and 70°. The third angle of the triangle measures

(A) 85° (B) 105° (C) 65° (D) 95° (E) 75°

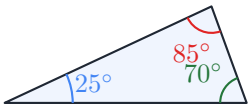
Detailed solution

Understanding the problem. Two angles of the triangle are known; we need the remaining one. We use the angle sum theorem.

Recall. The three angles of a triangle add up to 180°.

Step 1. Add the two known angles
$25° + 70° = 95°$.

Step 2. Subtract that sum from 180°
Third angle $= 180° - 95° = 85°$.



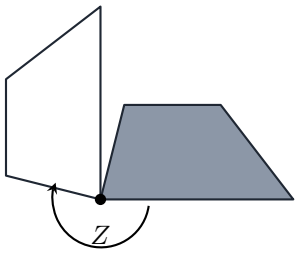
Common mistake. Option (D) 95° is the sum of the two given angles, not the third angle. Do not stop at step 1!

Check: $25° + 70° + 85° = 180°$. Correct.
So the answer is **(A)**.

Answer: (A) 85°

Problem 3

Rotating the shaded quadrilateral about point Z *clockwise* (in the direction of the arrow in the figure) produces the unshaded quadrilateral. The angle of rotation is approximately



(A) 180° (B) 270° (C) 360° (D) 45° (E) 135°

Detailed solution

Understanding the problem. The figure is rotated about Z clockwise. To find the angle, we follow *one side starting at* Z and see what angle it sweeps through.

Recall. In a rotation, every point turns through the same angle about the centre; side lengths do not change. A full turn is 360°; rotating α clockwise is the same as rotating $360° - \alpha$ counterclockwise.

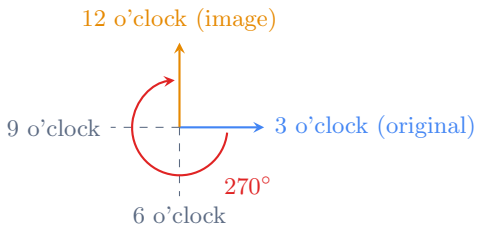
Step 1. Find the matching side

The shaded quadrilateral has its longest side starting at Z, horizontal and pointing *right* (the "3 o'clock" direction on a clock face). In the unshaded quadrilateral, the longest side starting at Z is the *vertical* side pointing *up* (the "12 o'clock" direction). Since a rotation keeps lengths, the horizontal side has become this vertical side.

Step 2. Measure the angle swept clockwise

Going clockwise from 3 o'clock to 12 o'clock, we pass 6 o'clock (down) and 9 o'clock (left) before reaching 12 o'clock: that is 3 right angles.

$$3 \cdot 90° = 270°.$$



Step 3. Check with a second side

The short slanted side of the shaded quadrilateral goes up from Z, leaning slightly right. After a further 270° clockwise turn (that is, 90° counterclockwise) it must point left, leaning slightly up – exactly the bottom side of the unshaded quadrilateral.

Common mistake. Rotating *counterclockwise* would need only 90°, but the problem asks for clockwise (the direction of the arrow), and 90° is not among the options. Do not mix up the two directions.

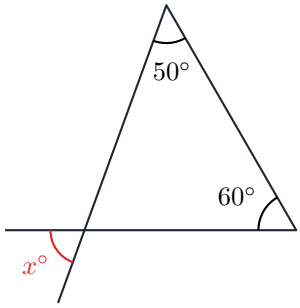
Check: $270° + 90° = 360°$: rotating $270°$ clockwise and rotating $90°$ counterclockwise give the same result, and a $90°$ counterclockwise turn clearly takes a side pointing "right" to a side pointing "up".
So the answer is **(B)**.

Answer: (B) $270°$

Note for parents. Take a clock (or draw a clock face) and let your child turn the minute hand clockwise from the 3 to the 12: the hand passes 9 "five-minute steps" = 45 minutes, i.e. $\frac{3}{4}$ of a turn $= \frac{3}{4} \cdot 360° = 270°$. This activity helps your child feel the difference between the two directions of rotation.

Problem 4

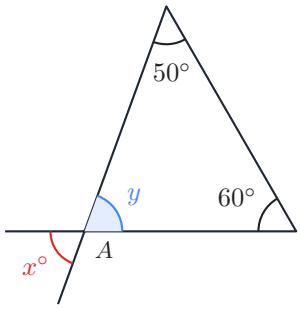
In the figure, two sides of the triangle are extended as shown. The value of x is



(A) 110 (B) 50 (C) 10 (D) 60 (E) 70

Detailed solution

Understanding the problem. The figure shows a triangle with two angles of $50°$ and $60°$. At the third vertex (bottom left), the base and a side are extended outwards, forming the angle $x°$. We first find the interior angle at that vertex, then connect it to x.



Step 1. Find the interior angle y at vertex A

Recall. The three angles of a triangle add up to 180°.

Let y be the interior angle of the triangle at vertex A:

$$y = 180° - (50° + 60°) = 180° - 110° = 70°.$$

Step 2. Spot the pair of vertically opposite angles
Angle $x°$ is formed by the extension of the base (to the left) and the extension of the side (downwards). These are exactly the extensions of the two arms of angle y. So $x°$ and y are **vertically opposite angles**.

Recall. Vertically opposite angles are equal.

Therefore $x° = y = 70°$, i.e. $x = 70$.

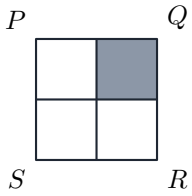
Common mistake. Option (A) 110 is what you get by confusing x with the angle *supplementary* to y (the angle right next to it, above the extended base). Look carefully: angle x sits *opposite* the interior angle across vertex A.

Check: The angle supplementary to y is $180° - 70° = 110°$; angle x is in turn supplementary to that $110°$ angle, so $x° = 180° - 110° = 70°$. Both ways give the same result.
So the answer is **(E)**.

Answer: (E) 70

Problem 5

Square $PQRS$ is divided into four equal smaller squares as shown. What percentage of the area of square $PQRS$ is shaded?



(A) 33% (B) 40% (C) 25% (D) 50% (E) 15%

Detailed solution

Understanding the problem. The big square is divided into 4 *equal* parts, and 1 part is shaded. We convert this ratio to a percentage.

Recall. If a figure is divided into n equal parts and m parts are shaded, the shaded part is $\frac{m}{n}$ of the figure, i.e. $\frac{m}{n} \cdot 100\%$.

Step 1. Write the fraction
The shaded part is 1 of 4 equal small squares: it is $\frac{1}{4}$ of square $PQRS$.

Step 2. Convert to a percentage
$\frac{1}{4} = \frac{25}{100} = 25\%$.

Check: If $PQRS$ has side 2 cm, the big area is 4 cm^2, and the shaded square has side 1 cm and area 1 cm^2; $1 \div 4 = 0.25 = 25\%$.

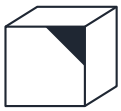
Common mistake. Option (A) 33% comes from comparing the shaded part with the *unshaded* part (1 to 3). A percentage must compare with the *whole* figure.

So the answer is **(C)**.

Answer: (C) 25%

Problem 6

A cube has part of *exactly one face* painted black as shown (the paint is a right triangle at a corner of the face, whose two legs are half the edge of the cube). The other five faces are not painted. After the cube is rolled, which of the following could be the same cube?



(A)  **(B)**  **(C)**  **(D)**  **(E)** 

Detailed solution

Understanding the problem. The cube has only *one* painted face, and that face carries only *one* black triangle. We describe that triangle very precisely and compare it with each option.

Recall. When a cube is rolled, each face is carried along unchanged: the pattern on a face may move or be turned by 90°, 180°, 270°, but it does *not* change size, is *not* mirrored and does *not* gain extra paint.

Step 1. Describe the original pattern
Three features of the pattern:

- there is only **one** black triangle;
- the triangle has its **right angle exactly at a corner** of the square face;
- its two legs are **half the edge** of the face (the triangle covers $\frac{1}{8}$ of the face).

Step 2. Examine each option

- (A): the black triangle covers *half* the face (its legs are whole edges) – too big, rejected.
- (B): one black triangle, right angle at the bottom-right corner of the face, legs half an edge – matches all three features.

- (C): one triangle of the right size, but its right angle is at the *midpoint* of the bottom edge, not at a corner of the face – rejected.
- (D) and (E): *two* black triangles, more paint than the original cube – rejected.

Step 3. Check that (B) can really be obtained
Pattern (B) is the original pattern *turned* 90° *clockwise* within the plane of that face: the top-right corner becomes the bottom-right corner. Place the cube on the table with the painted face towards you and turn the cube a quarter turn about the axis perpendicular to that face (a few rolls back and forth achieve this): we obtain exactly picture (B).

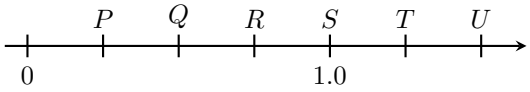
ThinkAlgo Tip. To check, cut out a paper square, colour a triangle in one corner and turn the paper: the triangle can only "travel" around the four corners, never sit in the middle of an edge as in (C).

So the answer is **(B)**.

Answer: (B)

Problem 7

On a number line the ticks are equally spaced. The numbers P, Q, R, S, T, U are marked as shown. Which of the following is closest to the value of $R \div U$?



(A) 0.25 (B) 0.50 (C) 0.75 (D) 1.25 (E) 1.50

Detailed solution

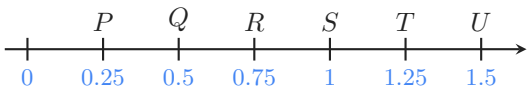
Understanding the problem. The two ticks with known values are 0 and 1.0 (tick S). From them we find the value of each gap, and then R and U.

Recall. On a number line with equal spacing: value of one gap = (difference of the two known numbers) ÷ (number of *gaps* between them).

Step 1. Count the gaps from 0 to 1.0
From tick 0 to tick S (at 1.0) the gaps are: $0 \to P$, $P \to Q$, $Q \to R$, $R \to S$, i.e. 4 gaps.

Step 2. Find the value of one gap
$1.0 \div 4 = 0.25$.

Step 3. Find R and U
R is three gaps from 0: $R = 3 \cdot 0.25 = 0.75$. U is six gaps from 0: $U = 6 \cdot 0.25 = 1.5$.



Step 4. Compute the quotient

$R \div U = 0.75 \div 1.5 = \dfrac{75}{150} = \dfrac{1}{2} = 0.5.$

Quick view: R is three gaps from 0 and U is six gaps from 0, so U is twice R and $R \div U = \frac{1}{2}$ – no need to know the size of a gap!

Common mistake. Option (C) 0.75 is the value of R, before dividing by U. Also, counting 5 *ticks* from 0 to S instead of 4 *gaps* gives a wrong value for each gap.

Check: $1.5 \cdot 0.5 = 0.75 = R$. Correct.

So the answer is **(B)**.

Answer: (B) 0.50

Problem 8

One angle of an isosceles triangle measures 50°. The other two angles of this triangle could be

(A) 50° and 90° (B) 40° and 50° (C) 50° and 80°

(D) 30° and 100° (E) 60° and 70°

Detailed solution

Understanding the problem. The problem does not say whether the 50° angle is the apex angle or a base angle, so we consider both cases and then compare with the options.

Recall. An isosceles triangle has two equal base angles; the three angles of a triangle add up to 180°.

Step 1. Case: 50° is the apex angle

The base angles are equal and add up to $180° - 50° = 130°$, so each is $130° \div 2 = 65°$. The other two angles: 65° and 65° (not among the options).

Step 2. Case: 50° is a base angle

The other base angle is also 50°. Apex angle: $180° - 50° - 50° = 80°$. The other two angles: 50° and 80° – this is option (C).

Step 3. Check why the other options are wrong

Each triple of angles must add up to 180° *and* contain two equal angles:

Option	Three angles	Sum	Two equal angles?
(A)	50°, 50°, 90°	190°	yes, but wrong sum
(B)	50°, 40°, 50°	140°	yes, but wrong sum
(C)	50°, 50°, 80°	180°	yes
(D)	50°, 30°, 100°	180°	no
(E)	50°, 60°, 70°	180°	no

Only (C) satisfies both conditions.

Check: $50° + 50° + 80° = 180°$ and two angles are equal (50°).

So the answer is **(C)**.

Answer: (C) 50° and 80°

Problem 9

Minh measures the length of his desk as 3.5 m. How many centimetres is that?

(A) 350 (B) 30.5 (C) 3.05 (D) 3.5 (E) 305

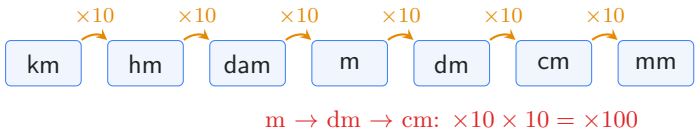
Detailed solution

Understanding the problem. We convert from metres (a larger unit) to centimetres (a smaller unit), so we *multiply*.

> **Recall.** 1 m = 10 dm = 100 cm. From a larger unit to a smaller unit, multiply; from a smaller unit to a larger unit, divide.

Step 1. Write the multiplication
$3.5 \text{ m} = 3.5 \cdot 100 \text{ cm}$.

Step 2. Multiply by 100
Multiplying a decimal by 100 moves the decimal point 2 places to the right: $3.5 \to 35 \to 350$. So 3.5 m = 350 cm.



m → dm → cm: $\times 10 \times 10 = \times 100$

Another way to see it: 3.5 m = 3 m + 0.5 m = 300 cm + 50 cm = 350 cm (half a metre is 50 cm).

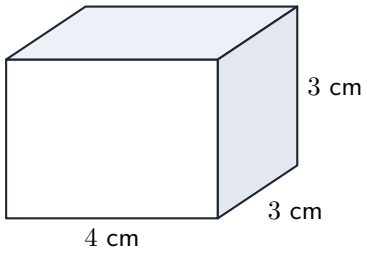
> **Common mistake.** Option (E) 305 comes from reading it as "3 metres 5 centimetres". But 0.5 m is *half a metre* = 50 cm, not 5 cm.

Check: $350 \div 100 = 3.5$. Correct.
So the answer is **(A)**.

Answer: (A) 350

Problem 10

An wants to build the solid rectangular box shown from small cubes measuring 1 cm × 1 cm × 1 cm. How many small cubes does An need?



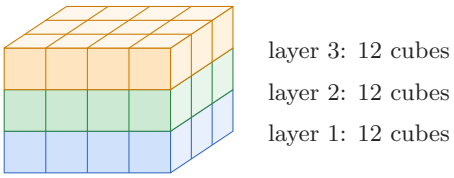
(A) 10 (B) 12 (C) 33 (D) 66 (E) 36

Detailed solution

Understanding the problem. The box has three dimensions: length 4 cm, height 3 cm, depth (width) 3 cm. The number of 1 cm^3 cubes needed is exactly the volume of the box in cm^3.

Recall. Volume of a rectangular box: $V = a \cdot b \cdot c$ (length × width × height).

Step 1. Count the cubes in one layer
We build horizontal layers. The bottom layer has a 4 cm × 3 cm base (4 cm long, 3 cm deep), so it has 3 rows of 4 cubes: $4 \cdot 3 = 12$ cubes.



Step 2. Count the layers
The height is 3 cm, so there are 3 layers stacked on top of each other.

Step 3. Multiply
Total number of cubes: $12 \cdot 3 = 36$ cubes. That is, $V = 4 \cdot 3 \cdot 3 = 36$ (cm^3).

Common mistake. Option (A) 10 is the *sum* $4 + 3 + 3$, and (C) 33 just puts digits together. The volume is the *product* of the three dimensions, not their sum.

Check: Counting another way: each vertical "slice" parallel to the front face has $4 \cdot 3 = 12$ cubes, and there are 3 slices front to back: $12 \cdot 3 = 36$.
So the answer is **(E)**.

Answer: (E) 36

Problem 11

A square garden has a perimeter of 36 m. The area of the garden, in m^2, is

(A) 24 (B) 81 (C) 36 (D) 1296 (E) 324

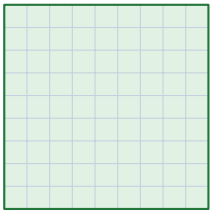
Detailed solution

Understanding the problem. We know the perimeter and want the area. The bridge between the two quantities is the *side length*: from the perimeter find the side, from the side find the area.

Recall. Square with side a: perimeter $P = 4 \cdot a$, area $S = a \cdot a$.

Step 1. Find the side of the square
A square has 4 equal sides, so $a = P \div 4 = 36 \div 4 = 9$ (m).

Step 2. Find the area
$S = 9 \cdot 9 = 81$ (m^2).



9 rows of 9 squares of 1 m^2:
$9 \cdot 9 = 81$ m^2

Common mistake.
- (C) 36: confusing perimeter with area.
- (D) $1296 = 36 \cdot 36$: squaring the perimeter instead of the side.
- (E) $324 = 36 \cdot 9$: multiplying the perimeter by the side.

Check: Side 9 m $\Rightarrow$ perimeter $4 \cdot 9 = 36$ m, as given.
So the answer is **(B)**.

Answer: (B) 81

ThinkAlgo Tip. For a square, *always* find the side first. A small table worth remembering: side 5 $\rightarrow$ perimeter 20, area 25; side 9 $\rightarrow$ perimeter 36, area 81; side 10 $\rightarrow$ perimeter 40, area 100. When the side doubles, the perimeter doubles but the area becomes *four* times as large.

Problem 12

Two isosceles triangles each have at least one angle of 70°. In the first triangle, each of the two remaining angles measures an even number of degrees. In the second triangle, each of the two remaining angles measures an odd number of degrees. Let S be the sum of the two equal angles of the first triangle and T the sum of the two equal angles of the second triangle. The value of $S + T$ is

(A) 280° (B) 250° (C) 220° (D) 200° (E) 300°

Detailed solution

Understanding the problem. This looks long, but it is really the familiar "isosceles triangle with an angle of 70°" problem. We list *all* isosceles triangles with a 70° angle, then use the even/odd conditions to decide which is the first triangle and which is the second.

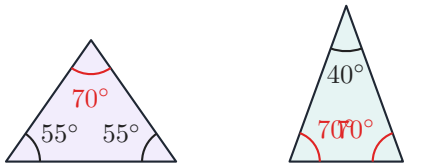
Recall. An isosceles triangle has two equal base angles; the three angles add up to 180°. A given angle may be the apex angle or a base angle – both must be considered.

Step 1. List the isosceles triangles with a 70° angle

Case 1: 70° *is the apex angle.* Each base angle: $(180° - 70°) \div 2 = 110° \div 2 = 55°$. Triangle: 70°, 55°, 55°.

Case 2: 70° *is a base angle.* The other base angle is also 70°, and the apex angle is $180° - 2 \cdot 70° = 40°$. Triangle: 70°, 70°, 40°.

There are only these two triangles.



70° at the apex 70° at the base

Step 2. Identify the first triangle (both remaining angles even)

Remove one 70° angle:

- triangle $70°, 55°, 55°$ leaves 55° and 55° – both *odd*;
- triangle $70°, 70°, 40°$ leaves 70° and 40° – both *even*.

So the first triangle is 70°, 70°, 40° and the second is 70°, 55°, 55°.

Step 3. Find S and T

The first triangle has equal angles 70° and 70°: $S = 70° + 70° = 140°$. The second triangle has equal angles 55° and 55°: $T = 55° + 55° = 110°$.

Step 4. Add

$S + T = 140° + 110° = 250°$.

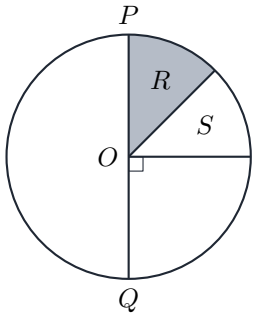
Common mistake. Read carefully: S and T are the sums of the *two equal angles*, not the sums of the two "remaining" angles (so for the first triangle we take $70° + 70°$, not $70° + 40°$).

Check: $70 + 70 + 40 = 180$ and $70 + 55 + 55 = 180$. Both triangles are valid; $140 + 110 = 250$. So the answer is **(B)**.

Answer: (B) 250°

Problem 13

On the circular spinner with centre O shown, PQ is a diameter, the horizontal radius is perpendicular to PQ at O, and the sectors R and S have equal areas. About what percentage of spins will stop in the shaded region?



(A) 50% (B) 22.5% (C) 25% (D) 45% (E) 12.5%

Detailed solution

Understanding the problem. The pointer is equally likely to stop in any direction, so the fraction of spins stopping in the shaded region equals the fraction of the area (also the fraction of the central angle) of that region compared with the whole circle. We need the central angle of sector R.

Recall. One turn around the centre is 360°. A sector with central angle $n°$ is $\frac{n}{360}$ of the area of the circle.

Step 1. Angle of R and S together
Radius OP points up and the horizontal radius points right; these two radii are perpendicular. So R and S together form a quarter of the circle, with central angle 90°.

Step 2. Angle of R alone
R and S have equal areas, so their central angles are equal: each is $90° \div 2 = 45°$.

Step 3. Convert to a percentage

$$\frac{45}{360} = \frac{1}{8} = 0.125 = 12.5\%.$$

Why is it true? Why does "equal areas" imply "equal angles"? The area of a sector is $\frac{n}{360}$ of the area of the circle, proportional to the angle n. Two sectors of the same circle with equal areas satisfy $\frac{n_1}{360} = \frac{n_2}{360}$, so $n_1 = n_2$. □

Common mistake. Option (C) 25% is the share of R and S *together*; (B) 22.5% comes from confusing 45° with 45% and halving. Only R is shaded, and the angle must be divided by 360.

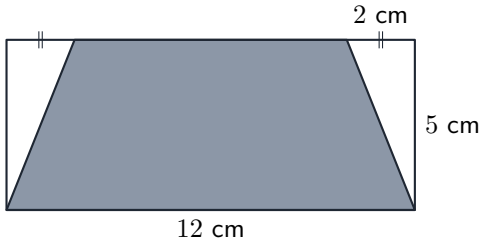
Check: The shaded part is half of a quarter circle, i.e. $\frac{1}{2} \cdot \frac{1}{4} = \frac{1}{8} = 12.5\%$.
So the answer is **(E)**.

Answer: (E) 12.5%

Note for parents. Fold a paper circle in half, then in quarters, then in eighths: each fold halves the smallest piece. Unfolded, your child will see 8 equal sectors, each with an angle of 45° and a share of 12.5% – exactly the region R in this problem.

Problem 14

In the rectangle below (12 cm long, 5 cm wide), the two segments marked alike on the top side are each 2 cm long. The area of the shaded region is



(A) 60 cm^2 (B) 20 cm^2 (C) 30 cm^2 (D) 40 cm^2 (E) 50 cm^2

Detailed solution

Understanding the problem. The shaded region is a trapezoid: its longer parallel side is the bottom side of the rectangle, and its shorter parallel side lies on the top side. The two segments at the ends of the top side are marked alike, which means they are *equal*, each 2 cm. We solve it in two ways to check ourselves.

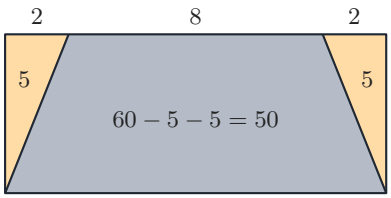
Step 1. Method 1 – rectangle minus two triangles

Recall. $S_{\text{rectangle}} = \text{length} \times \text{width}$; $S_{\text{right triangle}} = \dfrac{\text{product of the two legs}}{2}$.

Area of the rectangle: $12 \cdot 5 = 60$ (cm^2).
The two unshaded parts are right triangles in the corners; each has legs 2 cm (on the top side) and 5 cm (a side of the rectangle):

$$S_{\text{one triangle}} = \frac{2 \cdot 5}{2} = 5 \text{ (cm}^2\text{)}.$$

Shaded area: $60 - 5 - 5 = 50$ (cm^2).



Step 2. Method 2 – the trapezoid formula

Recall. $S_{\text{trapezoid}} = \dfrac{(a+b)\cdot h}{2}$.

Longer parallel side $a = 12$ cm; shorter parallel side $b = 12 - 2 - 2 = 8$ cm; height $h = 5$ cm (the distance between the top and bottom sides of the rectangle).

$$S = \frac{(12+8)\cdot 5}{2} = \frac{20\cdot 5}{2} = \frac{100}{2} = 50 \text{ (cm}^2\text{)}.$$

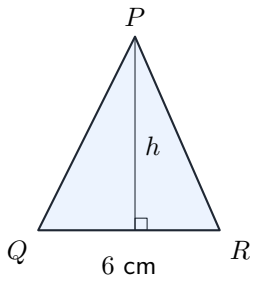
Common mistake. Option (A) 60 cm^2 is the area of the *whole* rectangle. Also remember that the height of the trapezoid is 5 cm (the perpendicular distance), not the length of a slanted side.

Check: Both methods give 50 cm^2.
So the answer is **(E)**.

Answer: (E) 50 cm^2

Problem 15

Triangle PQR has area 27 cm^2 and base $QR = 6$ cm. The height h of triangle PQR (to the base QR) is



(A) 9 cm (B) 18 cm (C) 4.5 cm (D) 2.25 cm (E) 7 cm

Detailed solution

Understanding the problem. We know the area and the base and need the height: this is the triangle area formula "worked backwards".

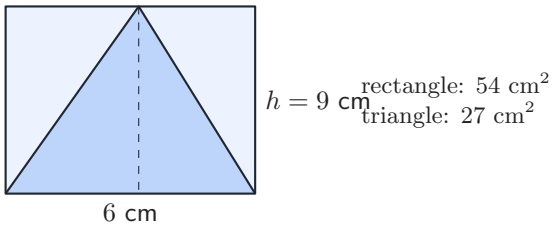
Recall. $S_{\text{triangle}} = \dfrac{a\cdot h}{2}$, where a is the base and h the height to that base. Hence $h = \dfrac{2\cdot S}{a}$.

Step 1. Substitute into the formula
$27 = \dfrac{6 \cdot h}{2}$.

Step 2. Get rid of the division by 2
Multiply both sides by 2: $6 \cdot h = 54$.

Step 3. Find h
$h = 54 \div 6 = 9$ (cm).

Why is it true? Why double the area? The triangle is half of the rectangle with the same base 6 cm and the same height h. That rectangle has area $2 \cdot 27 = 54$ cm^2 and one side 6 cm, so its other side (which is h) is $54 \div 6 = 9$ cm. □



Common mistake. Option (C) 4.5 cm is the result of $27 \div 6$ – forgetting that the area of a triangle is only *half* of base times height.

Check: $\dfrac{6 \cdot 9}{2} = \dfrac{54}{2} = 27$ (cm^2). Correct.
So the answer is **(A)**.

Answer: (A) 9 cm

Problem 16

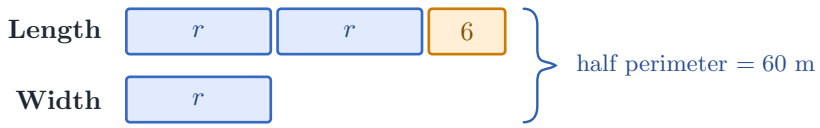
A rectangular garden has a length that is 6 m more than *twice* its width. The perimeter of the garden is 120 m. How many metres wide is the garden?

(A) 8 m (B) 18 m (C) 27 m (D) 38 m (E) 22 m

Detailed solution

Understanding the problem. We know two things: (1) length = twice the width, plus 6 m; (2) perimeter = 120 m. We must find the width.

Step 1. Call the width r (m) and express the length in terms of r
"Twice the width" is $2r$; "6 m more than twice the width" means the length is $2r + 6$ (m). We draw a bar model:



Recall. Perimeter of a rectangle = (length + width) × 2. Therefore length + width = perimeter ÷2 (called the *half perimeter*).

Step 2. Compute the half perimeter
$120 \div 2 = 60$ (m). So length + width = 60 m.

Step 3. Write and solve the equation
From the bar model: length + width consists of 3 bars r and one bar of 6 m:

$$(2r + 6) + r = 60 \;\Rightarrow\; 3r + 6 = 60 \;\Rightarrow\; 3r = 60 - 6 = 54 \;\Rightarrow\; r = 54 \div 3 = 18.$$

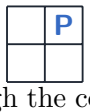
The width is 18 m and the length is $2 \cdot 18 + 6 = 42$ (m).

Common mistake. Many students forget to halve the perimeter and write $3r + 6 = 120$, which gives $r = 38$ – exactly the "trap" option (D). Remember: the perimeter contains *two* lengths and *two* widths.

Check: Length 42 m, width 18 m: $42 = 2 \cdot 18 + 6$ (correct) and the perimeter is $(42 + 18) \cdot 2 = 60 \cdot 2 = 120$ m (correct).
A width of 18 m is option **(B)**.

Answer: (B) 18 m

Problem 17

The letter P is written in a 2×2 grid of squares as shown: 

. A sequence of rotations about the centre of the grid and reflections in the two lines through the centre of the grid (the horizontal line and the vertical line) is carried out, and the result is: 

.

If *exactly the same sequence of transformations* is applied to the grid 

, which picture do we get?

(A) 

(B) 

(C) 

(D) 

(E) 

Detailed solution

Understanding the problem. We do not know which individual steps the sequence contains, but we know its *final effect* on the letter P. Idea: imagine the grid is a square *card* made of clear plastic with the letter drawn on it. Every rotation or reflection just turns or flips the card and puts it back in the same place. The whole sequence is therefore also just *one way of putting the card back*. We read off that way from the picture of P, and then apply it to A.

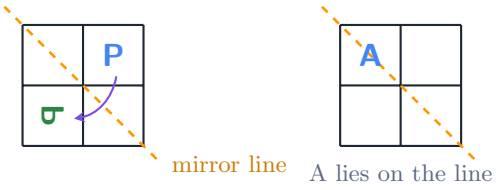
Step 1. See where the letter P went
The letter P in the *top-right* cell has moved to the *bottom-left* cell. At the same time, the straight stroke of P (its "stem"), which was vertical, is now horizontal at the top; the round "bowl", which was to the right of the stem, is now underneath it.

Step 2. Find out how the card was put back

There are two ways of putting the card back that move the top-right cell to the bottom-left cell:

- a 180° rotation: then P would be upside down (its stem still *vertical*, looking like a "d");
- a reflection in the diagonal joining the top-left corner to the bottom-right corner (the dashed line in the picture below): then the vertical stem of P becomes *horizontal*.

In the given picture the stem is horizontal, so the sequence acts like the **reflection in the top-left – bottom-right diagonal**.



Recall. A sequence of rotations about the centre and reflections in the horizontal and vertical lines can have the same effect as a reflection in a diagonal. For example: *a 90° clockwise rotation followed by a reflection in the vertical line*. Check with P: the 90° rotation takes P from the top-right cell down to the bottom-right cell, the reflection in the vertical line takes it to the bottom-left cell, and the stem of P is horizontal – exactly as in the given picture.

Step 3. Apply it to the letter A

The top-left cell lies *right on* the mirror line (the diagonal passes through it), so the letter A *stays in the top-left cell*. This diagonal reflection turns the direction "up" into the direction "left". The tip of A, which pointed up, now points left. The letter A is symmetric left-to-right, so we need not worry about it being "mirrored".

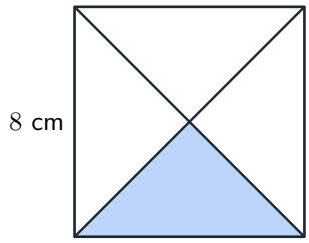
Common mistake. Do not look only at the position of the cell! P moved from the top-right to the bottom-left cell, but that does *not* mean A also "moves diagonally down". First find the transformation of *the whole card*, then apply it.

Check: Use the concrete sequence "90° clockwise rotation, then reflection in the vertical line": the rotation moves A from the top-left to the top-right cell with its tip pointing right; the reflection in the vertical line brings A back to the top-left cell with its tip pointing left. This is picture (B).

Answer: (B) A in the top-left cell, tip pointing left

Problem 18

In a square with side 8 cm, the two diagonals are drawn. What is the area of the shaded region?



(A) 4 cm^2 (B) 8 cm^2 (C) 16 cm^2 (D) 56 cm^2 (E) 64 cm^2

Detailed solution

Understanding the problem. The two diagonals divide the square into 4 triangles. The shaded region is one of these four triangles. We need its area.

Step 1. Compute the area of the square

Recall. The area of a square with side a is $a \times a$.

$A_{\text{square}} = 8 \times 8 = 64$ (cm^2).

Step 2. Show that the four triangles have equal areas

Why is it true? The two diagonals of a square are equal and bisect each other at a point O, so O is the same distance from all four vertices. Rotating the square 90° about O moves each triangle exactly onto the next one. So the four triangles are identical, and each is $\frac{1}{4}$ of the square. □

Step 3. Compute the shaded area

$A = 64 \div 4 = 16$ (cm^2).

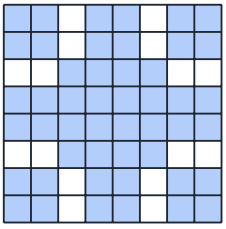
Check: Direct calculation: the shaded triangle has the bottom side as its base, 8 cm, and its height is the distance from the centre O to that side, which is half the side, 4 cm. Area $= \dfrac{8 \times 4}{2} = 16$ cm^2. It matches.

Option **(C)**.

Answer: (C) 16 cm^2

Problem 19

The grid below is made of identical small squares. What percentage of the area of the whole grid is shaded?



(A) 84% (B) 80% (C) 90% (D) 75% (E) 66%

Detailed solution

Understanding the problem. To find a percentage we need to know how many squares there are in total and how many are shaded. Counting the *white* squares is quicker because there are fewer of them.

Step 1. Count all the squares

The grid has 8 rows of 8 squares: $8 \times 8 = 64$ squares.

Step 2. Count the white squares

The white squares form 8 "pairs" of two adjacent squares, placed symmetrically on the four sides:

- top two rows: columns 3 and 6 have a white square in both rows: 4 squares;
- bottom two rows: likewise, 4 squares;
- row 3 (from the top): 2 white squares at the left edge and 2 at the right edge: 4 squares;
- row 6: likewise, 4 squares.

Total number of white squares: $4 + 4 + 4 + 4 = 16$.

Step 3. Count the shaded squares and convert to a percentage

Number of shaded squares: $64 - 16 = 48$.

Recall. The percentage that a is of b equals $\dfrac{a}{b} \times 100\%$.

$$\frac{48}{64} = \frac{3}{4} = \frac{75}{100} = 75\%.$$

ThinkAlgo Tip. You do not even need to count everything: the white squares make up $\dfrac{16}{64} = \dfrac{1}{4} = 25\%$, so the shaded squares make up $100\% - 25\% = 75\%$.

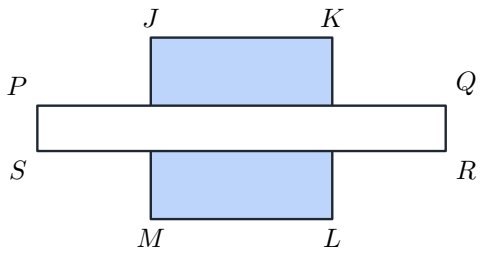
Check: 75% of 64 is $64 \times \dfrac{3}{4} = 48$ squares, and $48 + 16 = 64$. Correct.

Option **(D)**.

Answer: (D) 75%

Problem 20

$JKLM$ is a square and $PQRS$ is a rectangle; they overlap as shown in the diagram. Given that JK is parallel to PQ, $JK = 8$ and $PS = 2$, find the total area of the shaded regions.



(A) 32 (B) 16 (C) 56 (D) 48 (E) 62

Detailed solution

Understanding the problem. The shaded part is the part of the square that is *not* covered by the rectangular strip. So: shaded area = area of the square − area of the overlap of the two figures.

Step 1. Area of the square $JKLM$

Recall. Area of a square = side × side; area of a rectangle = length × width.

$A_{JKLM} = 8 \times 8 = 64$.

Step 2. Identify the overlap

Since $JK \parallel PQ$, the sides of the strip $PQRS$ are parallel to the side JK of the square, i.e. the strip runs "horizontally" across the square. The strip is longer than the square, so it crosses the whole width of the square. The overlap is a rectangle with:

- length equal to the side of the square: 8;
- width equal to the width of the strip: $PS = 2$.

Area of the overlap: $8 \times 2 = 16$.

Step 3. Compute the shaded area

$64 - 16 = 48$.

Common mistake. We do not need to know how high or low the strip is, or how big each shaded piece is: their total is always $64 - 16 = 48$. Many students only compute $8 \times 2 = 16$ (option B) – that is the *covered* part, not the shaded part.

Check: Suppose the strip is 3 below the top side JK; the top piece has area $8 \times 3 = 24$, the bottom piece has height $8 - 3 - 2 = 3$, area 24. Total 48. Correct.
Option **(D)**.

Answer: (D) 48

Problem 21

A triangle can be built with side lengths 4, 5, 8, but *no* triangle can be built with side lengths 4, 5, 10. Using only the lengths 2, 3, 5, 7, 11 (each length may be used more than once), how many different triangles with *exactly two equal sides* can be built?

(A) 8 (B) 5 (C) 20 (D) 10 (E) 14

Detailed solution

Understanding the problem. A triangle with exactly two equal sides is isosceles but not equilateral: two legs of length a and a base of length b with $b \neq a$. We must count the pairs (a, b) taken from $\{2, 3, 5, 7, 11\}$ such that the three segments a, a, b really form a triangle.

> **Recall. Triangle inequality.** Three segments are the sides of a triangle exactly when the sum of any two of them is greater than the third. In practice it is enough to check that *the two shorter segments add up to more than the longest one.* Example: $4 + 5 = 9 > 8$, so it works; $4 + 5 = 9 < 10$, so it does not.

Step 1. Simplify the condition for an isosceles triangle a, a, b
The sums to check are $a + a > b$ and $a + b > a$. The second is always true (since $b > 0$). So we only need

$$2a > b \quad \text{(twice the leg is greater than the base).}$$

Step 2. Make a table by the leg a

Leg a	$2a$	Base $b \neq a$ with $b < 2a$	Number of triangles
2	4	3	1
3	6	2, 5	2
5	10	2, 3, 7	3
7	14	2, 3, 5, 11	4
11	22	2, 3, 5, 7	4

For instance, with $a = 3$: the base $b = 7$ does not work because $3 + 3 = 6 < 7$; $b = 11$ does not work either. With $a = 5$: $b = 11$ does not work because $5 + 5 = 10 < 11$.

Step 3. Add up
$1 + 2 + 3 + 4 + 4 = 14$ triangles.

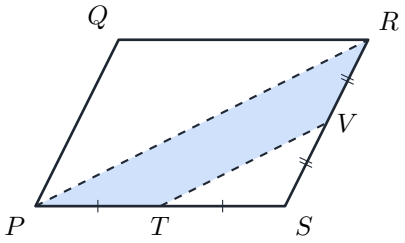
> **Common mistake.** Triangles such as $2, 2, 5$ or $3, 3, 7$ "look" possible, but in fact the two legs are too short to meet. Also, the triangle $5, 5, 5$ has *three* equal sides, so it is not counted.

Check: Count the other way: there are $5 \times 4 = 20$ pairs (a, b) with $b \neq a$. The rejected pairs (because $b \geq 2a$) are $(2, 5), (2, 7), (2, 11), (3, 7), (3, 11), (5, 11)$: 6 pairs. $20 - 6 = 14$. It matches. Option **(E)**.

Answer: (E) 14

Problem 22

Parallelogram $PQRS$ has area 40. Points T and V are the midpoints of sides PS and RS respectively. Find the area of quadrilateral $PRVT$.



(A) 10 (B) 12 (C) 15 (D) 16 (E) 18

Detailed solution

Understanding the problem. Quadrilateral $PRVT$ is what remains when the small triangle TVS at corner S is removed from triangle PRS. So we compute A_{PRS} and A_{TVS}.

Step 1. Area of triangle PRS

Recall. A diagonal divides a parallelogram into two identical triangles, so each triangle has half the area of the parallelogram.

$A_{PRS} = 40 \div 2 = 20$.

Step 2. Area of triangle TRS (join T to R)

Recall. Two triangles with *the same height* have areas in the same ratio as their bases. (Because the area of a triangle $= \frac{1}{2} \times$ base $\times$ height.)

Triangles TRS and PRS share the vertex R, and their bases TS and PS lie on the same line, so they have the same height from R. Since T is the midpoint of PS, $TS = \frac{1}{2}PS$. Therefore $A_{TRS} = \frac{1}{2} \cdot 20 = 10$.

Step 3. Area of triangle TVS

Triangles TVS and TRS share the vertex T, and their bases VS and RS lie on the line RS. Since V is the midpoint of RS, $VS = \frac{1}{2}RS$, so $A_{TVS} = \frac{1}{2} \cdot 10 = 5$.

Step 4. Area of quadrilateral $PRVT$

$A_{PRVT} = A_{PRS} - A_{TVS} = 20 - 5 = 15$.

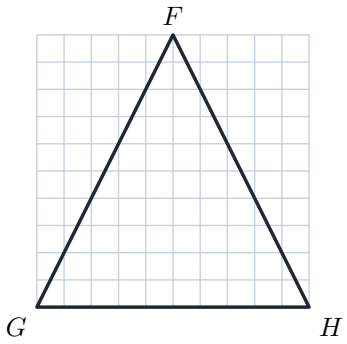
ThinkAlgo Tip. A "half – half" triangle at a corner has area $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$ of the big triangle. This is a very useful fact!

Check: Place the figure on a grid: $P(0,0)$, $S(6,0)$, $R(8,4)$, $Q(2,4)$ (area $6 \times 4 = 24$, not 40, but the ratios are the same). Then $T(3,0)$, $V(7,2)$; $A_{PRS} = 12$, $A_{TVS} = \frac{1}{2} \cdot 3 \cdot 2 = 3$, so $A_{PRVT} = 9 = \frac{3}{8} \cdot 24$. For the given figure: $\frac{3}{8} \cdot 40 = 15$. It matches.
Option **(C)**.

Answer: (C) 15

Problem 23 *

On a 10×10 grid of squares (each square has side 1), triangle FGH is drawn as shown. Point P can be any of the 41 points where gridlines cross *inside* (not on the sides of) triangle FGH. For each position of P we get exactly three triangles: FPG, GPH, HPF. Of all these 123 triangles, how many have an area *exactly half* the area of triangle FGH?



(A) 5 (B) 7 (C) 3 (D) 11 (E) 9

Detailed solution

Understanding the problem. Use coordinates: $G(0,0)$, $H(10,0)$, $F(5,10)$. Triangle FGH has base $GH = 10$ and height 10, so

$$A_{FGH} = \frac{10 \times 10}{2} = 50, \qquad \text{and half of it is } 25.$$

We count, for each type of triangle GPH, FPG, HPF in turn, the positions of P that give area 25.

Recall. Two triangles with *the same base* have areas in the same ratio as their heights. Points that are the same distance from a line (on the same side) lie on a line parallel to it.

Step 1. Triangle GPH

Triangle GPH has the same base GH as triangle FGH. For half the area, the height from P must be half of the height 10, i.e. 5. So P lies on the 5th horizontal gridline (counting from GH). On this line, side FG passes through the point with x-coordinate 2.5 and side FH through the point with x-coordinate 7.5 (the midpoints of those sides). The grid points *inside* have $x = 3, 4, 5, 6, 7$:

$$(3,5),\ (4,5),\ (5,5),\ (6,5),\ (7,5) \quad \Rightarrow \quad 5 \text{ triangles.}$$

Step 2. Triangle FPG

This time the common base is FG. Let $M(5,0)$ be the midpoint of GH and $N(7.5,5)$ the midpoint of FH.

Why is it true? Triangles FMG and FHG have the same height from F, and the base $GM = \frac{1}{2}GH$, so $A_{FMG} = \frac{1}{2}A_{FHG} = 25$. So point M is exactly half as far from FG as H is. The same holds for N (the midpoint of FH). Hence $A_{FPG} = 25$ exactly when P is the same distance from FG as M, i.e. when P lies on the line MN (parallel to FG). □

Going from G to F we move 5 right and 10 up, i.e. "1 right, 2 up". Line MN is parallel to FG, so from $M(5,0)$ we also go "1 right, 2 up": we meet $(6,2)$, $(7,4)$, and then $N(7.5,5)$ (not a grid point). The two points $(6,2)$, $(7,4)$ are inside the triangle $\Rightarrow$ 2 triangles.

Step 3. Triangle HPF

The figure is symmetric about the vertical line $x = 5$, so the result is the same as in step 2: $P = (4,2)$ or $(3,4) \Rightarrow$ 2 triangles.

Step 4. Add up

$5 + 2 + 2 = 9$ triangles.

Check: Check $P(6,2)$ with the "enclosing rectangle" method: the rectangle from $x = 0$ to 6 and $y = 0$ to 10 has area 60. Remove the three right triangles outside triangle FPG: $\frac{5\cdot 10}{2} = 25$, $\frac{1\cdot 8}{2} = 4$, $\frac{6\cdot 2}{2} = 6$. What remains is $60 - 35 = 25$, exactly half of 50.

Common mistake. Do not forget the two types FPG and HPF! If you only consider triangles with the horizontal base GH, you will wrongly choose option (A) 5.

Option **(E)**.

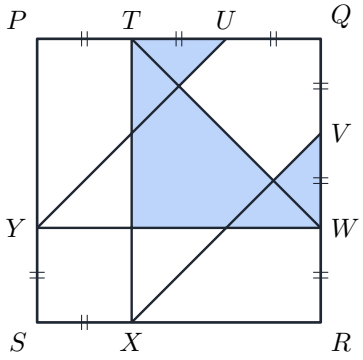
Answer: (E) 9

Problem 24

Points T, U, V, W, X, Y lie on the sides of square $PQRS$ as shown, with

$$PT = TU = UQ = QV = VW = WR = XS = SY.$$

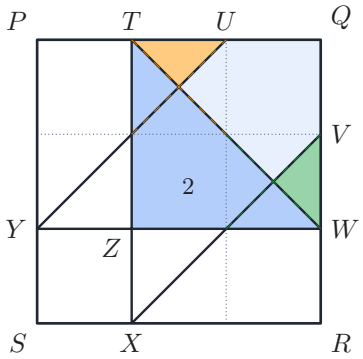
What fraction of the area of square $PQRS$ is shaded?



(A) $\frac{5}{18}$ (B) $\frac{1}{3}$ (C) $\frac{2}{9}$ (D) $\frac{1}{4}$ (E) $\frac{1}{6}$

Detailed solution

Understanding the problem. Each side of the square is divided into 3 equal parts. Let each part have length 1 unit; then the square has side 3 and area $3 \times 3 = 9$. A nice idea: draw extra lines dividing the square into a 3×3 grid of unit squares (dotted lines); then every line in the figure becomes very "tidy".



Step 1. Recognise the three pieces of the shaded region
Let Z be the point where TX meets YW. The shaded region consists of three non-overlapping pieces: the big triangle TZW (dark blue), the small triangle next to side TU (orange) and the small triangle next to side VW (green).

Step 2. Triangle TZW
$TQWZ$ is a 2×2 square (made of 4 unit squares) and TW is its diagonal.

> **Recall.** A diagonal divides a square (rectangle) into two triangles of equal area.

So $A_{TZW} = \frac{1}{2} \cdot 4 = 2$.

Step 3. The two small triangles
Look at the unit square whose top side is TU. Line YU goes "1 right, 1 up" through grid corners, so it contains one diagonal of this unit square; line TW goes "1 right, 1 down" from T, so it contains the other diagonal. The two diagonals of a square divide it into 4 equal triangles, and the orange triangle is one of them: area $\frac{1}{4} \cdot 1 = \frac{1}{4}$. In exactly the same way, the green triangle lies in the unit square whose right side is VW, cut by the two diagonals XV and TW: its area is also $\frac{1}{4}$.

Step 4. Compute the fraction
Shaded area: $2 + \frac{1}{4} + \frac{1}{4} = \frac{5}{2}$. The required fraction:

$$\frac{5}{2} \div 9 = \frac{5}{2} \cdot \frac{1}{9} = \frac{5}{18}.$$

> **Common mistake.** Do not "estimate by eye"! The shaded part looks close to $\frac{1}{3}$, but the exact calculation gives $\frac{5}{18}$ (a little less than $\frac{6}{18} = \frac{1}{3}$).

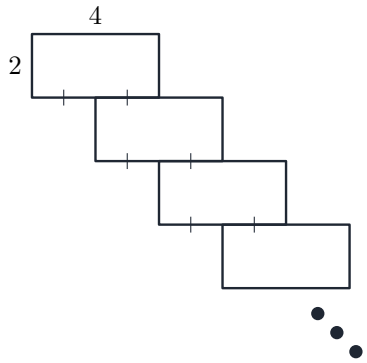
Check: Put $S(0,0)$ and side 3; computing the area of the shaded polygon by adding the triangles gives the same result $\frac{5}{2}$, and $\frac{5}{2} = \frac{5}{18} \cdot 9$. Correct.
Option **(A)**.

Answer: (A) $\frac{5}{18}$

Problem 25

Rectangles measuring 4×2 are arranged by the following rule: from the second rectangle on, the top-left vertex of each rectangle is placed exactly at the midpoint of the bottom side of the

rectangle just above it (as shown). When 10 rectangles in total are arranged in this way, what is the perimeter of the whole figure?



(A) 48 (B) 64 (C) 90 (D) 84 (E) 100

Detailed solution

Understanding the problem. Each new rectangle is shifted 2 to the right (half the length) and 2 down (exactly the width). The whole figure looks like a "staircase". We compute the perimeter in two ways.

Step 1. Method 1: count each kind of side

Vertical sides. Each rectangle has a left side and a right side of length 2, and both lie on the boundary (not covered by another rectangle). Total: $10 \times (2+2) = 40$.

Horizontal sides on top. The first rectangle shows its whole top side: 4. For each later rectangle, the left half of its top side is against the bottom side of the previous one, so only the right half shows: 2. Total: $4 + 9 \times 2 = 22$.

Horizontal sides at the bottom. Each of the first 9 rectangles shows only the left half of its bottom side: 2; the last one shows its whole bottom side: 4. Total: $9 \times 2 + 4 = 22$.

Perimeter: $40 + 22 + 22 = 84$.

Step 2. Method 2: "straighten" the staircase

Why is it true? Walking along the top-right boundary from the top-left corner of the first rectangle to the bottom-right corner of the last one, we only ever go right or down. The horizontal pieces add up to exactly the width of the whole figure, and the vertical pieces add up to exactly its height. The bottom-left boundary is the same. Hence the perimeter of the staircase *equals the perimeter of the smallest rectangle enclosing it.* □

Width: $4 + 9 \times 2 = 22$; height: $10 \times 2 = 20$. Perimeter: $(22 + 20) \times 2 = 84$.

Common mistake. Do not take $10\times$ the perimeter of one rectangle ($10 \times 12 = 120$): the pieces where two rectangles touch are *inside*, not on the boundary.

Check: Try 2 rectangles: method 1 gives $8 + (4+2) + (2+4) = 20$; the enclosing rectangle is 6×4, perimeter 20. With n rectangles the perimeter is $2\big((4 + 2(n-1)) + 2n\big) = 8n + 4$; $n = 10$ gives 84. It matches.

Option **(D)**.

Answer: (D) 84

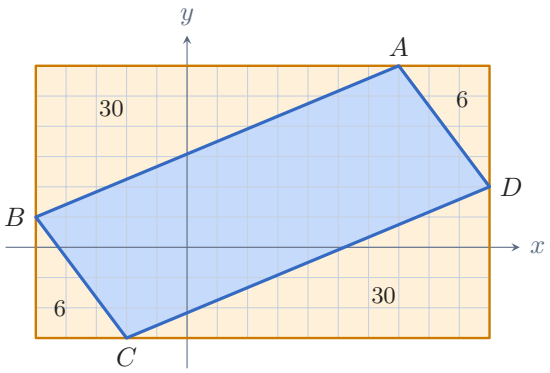
Problem 26

A quadrilateral has vertices $A(7,6)$, $B(-5,1)$, $C(-2,-3)$ and $D(10,2)$ in the coordinate plane. The area of the quadrilateral (in square units) is

(A) 60 (B) 63 (C) 67 (D) 70 (E) 72

Detailed solution

Understanding the problem. The quadrilateral has "slanted" sides, and there is no ready-made formula. A standard technique: *enclose the quadrilateral in a rectangle with sides parallel to the axes*, then subtract the right triangles in the four corners.



Step 1. The enclosing rectangle

The smallest x-coordinate is -5, the largest is 10; the smallest y-coordinate is -3, the largest is 6.

> **Recall.** The distance between two points on a number line is the larger number minus the smaller: from -5 to 10 it is $10-(-5)=15$.

The rectangle has length $10-(-5)=15$, width $6-(-3)=9$, and area $15\times 9=135$.

Step 2. The four corner right triangles

> **Recall.** Area of a right triangle $=\frac{1}{2}\times$ the product of the two legs.

- Top-left corner (between B and A): legs $7-(-5)=12$ and $6-1=5$: area $\frac{12\cdot 5}{2}=30$.
- Bottom-left corner (between B and C): legs $-2-(-5)=3$ and $1-(-3)=4$: area $\frac{3\cdot 4}{2}=6$.
- Bottom-right corner (between C and D): legs $10-(-2)=12$ and $2-(-3)=5$: area 30.
- Top-right corner (between D and A): legs $10-7=3$ and $6-2=4$: area 6.

Total: $30+6+30+6=72$.

Step 3. Area of the quadrilateral

$135-72=63$.

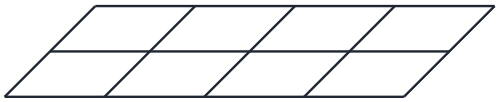
Common mistake. With negative coordinates it is easy to get a length wrong, e.g. writing $7-5=2$ instead of $7-(-5)=12$. Count squares on the picture to check.

Check: In fact $ABCD$ is a parallelogram ($A \to B$ and $D \to C$ are both "12 left, 5 down"). Triangle ABD has half the area of the quadrilateral; enclose it in a 15×5 rectangle (from $y=1$ to 6): $75 - 30 - \frac{15 \cdot 1}{2} - 6 = 31.5$. Doubling: 63. It matches.
Option **(B)**.

Answer: (B) 63

Problem 27

The figure below is made of 8 identical small parallelograms joined together. Including these 8 small parallelograms, how many parallelograms are there in the figure altogether?



(A) 29 (B) 30 (C) 26 (D) 27 (E) 28

Detailed solution

Understanding the problem. Every parallelogram in the figure has two sides on two *horizontal* lines and two sides on two *slanted* lines of the figure. We count systematically so as not to miss or repeat any.

Step 1. Count by height and width (in small cells)
Parallelograms 1 *row high* (there are 2 rows to choose from). A row has 4 cells side by side:

- 1 cell wide: 4; 2 cells wide: 3; 3 cells wide: 2; 4 cells wide: 1;
- each row gives $4+3+2+1=10$, and the two rows give 20.

Parallelograms 2 *rows high*: likewise $4+3+2+1=10$.
Total: $20+10=30$.

Step 2. A faster count: choose the lines

Why is it true? A parallelogram is completely determined by choosing 2 of the 3 horizontal lines (top and bottom sides) and 2 of the 5 slanted lines (left and right sides). Conversely, each such choice gives exactly one parallelogram. □

Recall. The number of ways to choose 2 lines out of k lines is $\dfrac{k \times (k-1)}{2}$ (each pair is counted twice if we pick a "first line" and a "second line").

Choose 2 of the 3 horizontal lines: $\frac{3 \cdot 2}{2} = 3$ ways. Choose 2 of the 5 slanted lines: $\frac{5 \cdot 4}{2} = 10$ ways.
Number of parallelograms: $3 \times 10 = 30$.

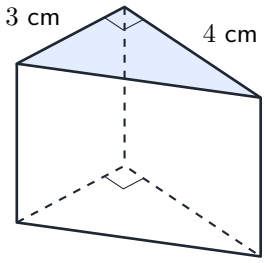
Common mistake. Students often forget the parallelograms 2 rows high, or the biggest one (the whole figure).

Check: The two counting methods give the same result, 30.
Option **(B)**.

Answer: (B) 30

Problem 28

A right triangular prism has volume 120 cm^3. Its base is a right triangle whose two legs are 3 cm and 4 cm (as shown). What is the height of the prism, in centimetres?



(A) 12 (B) 20 (C) 10 (D) 16 (E) 8

Detailed solution

Understanding the problem. We know the volume and the base (a right triangle). We must find the height of the prism, i.e. the length of a lateral edge (joining the top base to the bottom base).

Recall. Volume of a right prism = area of the base × height: $V = A_{\text{base}} \cdot h$. Therefore $h = V \div A_{\text{base}}$.

Step 1. Area of the base
The base is a right triangle; taking one leg as the base, the other leg is the height:

$$A_{\text{base}} = \frac{3 \times 4}{2} = 6 \text{ (cm}^2\text{)}.$$

Step 2. Height of the prism
$h = 120 \div 6 = 20$ (cm).

Common mistake. Do not confuse "the height of the prism" (the distance between the two bases) with a height of the base triangle. And do not forget to divide by 2 for the area of the triangle: using $3 \times 4 = 12$ gives 10 – the "trap" option (C).

Check: $6 \times 20 = 120$ cm^3. Correct.
Option **(B)**.

Answer: (B) 20 cm

Problem 29 *

Small cubes with edge 1 are used to build a rectangular box of size $8 \times 8 \times n$ (n a positive integer). Let A be the surface area of the box and let B be the total surface area of all the small cubes (computed for each cube separately and then added up). Find the sum of all values of n for which $\dfrac{B}{A}$ is an integer.

(A) 86 (B) 90 (C) 70 (D) 78 (E) 96

Detailed solution

Understanding the problem. We write A and B in terms of n, form the ratio $\dfrac{B}{A}$, and find the values of n that make it an integer.

Step 1. Compute A

> **Recall.** A rectangular box of size $a \times b \times c$ has 6 faces forming three pairs of equal faces, so its surface area is $2(ab + bc + ca)$.

$A = 2(8 \cdot 8 + 8 \cdot n + 8 \cdot n) = 2(64 + 16n) = 128 + 32n$.

Step 2. Compute B

Number of small cubes: $8 \cdot 8 \cdot n = 64n$. Each cube has 6 faces of area 1, so $B = 64n \cdot 6 = 384n$.

Step 3. Simplify the ratio

$$\frac{B}{A} = \frac{384n}{128 + 32n} = \frac{32 \cdot 12n}{32 \cdot (4 + n)} = \frac{12n}{n + 4}.$$

We "split off the whole part": since $12n = 12(n + 4) - 48$,

$$\frac{12n}{n + 4} = \frac{12(n + 4) - 48}{n + 4} = 12 - \frac{48}{n + 4}.$$

Step 4. When is the ratio an integer?

$12 - \dfrac{48}{n + 4}$ is an integer exactly when $\dfrac{48}{n + 4}$ is an integer, i.e. when $n + 4$ is a *divisor* of 48.

> **Recall.** The positive divisors of 48: $1, 2, 3, 4, 6, 8, 12, 16, 24, 48$.

Since $n \geq 1$, we have $n + 4 \geq 5$, so we take only the divisors $6, 8, 12, 16, 24, 48$, which give

$$n = 2,\ 4,\ 8,\ 12,\ 20,\ 44.$$

Step 5. Add them up

$2 + 4 + 8 + 12 + 20 + 44 = 90$.

n	2	4	8	12	20	44
$B/A = \dfrac{12n}{n+4}$	$\frac{24}{6} = 4$	$\frac{48}{8} = 6$	$\frac{96}{12} = 8$	$\frac{144}{16} = 9$	$\frac{240}{24} = 10$	$\frac{528}{48} = 11$

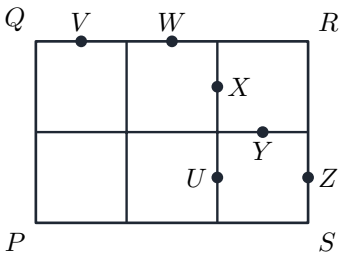
Common mistake. Do not forget that $n+4$ can be 48 (i.e. $n=44$), a rather large value. And remember to reject the divisors $1, 2, 3, 4$, because then $n \leq 0$.

Check: The table shows that all 6 values give an integer ratio. Try another value, $n=1$: $\frac{12}{5}$ is not an integer. Correct.
Option **(B)**.

Answer: (B) 90

Problem 30

Rectangle $PQRS$ is made up of six identical squares. Points U, V, W, X, Y, Z are midpoints of sides of the squares, as shown. Which of the following triangles has the greatest area?



(A) PVU (B) PXZ (C) PVX (D) PYS (E) PQW

Detailed solution

Understanding the problem. We need to compute (or compare) the areas of 5 triangles. To avoid fractions, let each small square have side 2; then all the midpoints have whole-number coordinates. Put $P(0,0)$, $S(6,0)$, $R(6,4)$, $Q(0,4)$; then $V(1,4)$, $W(3,4)$, $X(4,3)$, $Y(5,2)$, $U(4,1)$, $Z(6,1)$.

Recall. Area of a triangle $= \frac{1}{2} \times$ base $\times$ height. For a "slanted" triangle, enclose it in a rectangle with sides along the grid and subtract the corner right triangles (as in Problem 26).

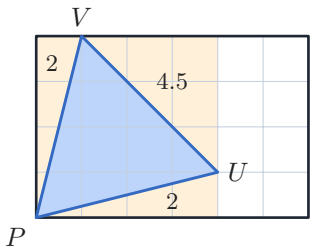
Step 1. Two triangles with a side on a side of the rectangle

- PYS: base $PS = 6$, height = distance from Y to PS, which is 2: $A = \frac{6 \cdot 2}{2} = 6$.
- PQW: base $PQ = 4$, height = distance from W to PQ, which is 3: $A = \frac{4 \cdot 3}{2} = 6$.

Step 2. Three "slanted" triangles

- PVU: enclosing rectangle $4 \times 4 = 16$; subtract $\frac{1 \cdot 4}{2} = 2$, $\frac{3 \cdot 3}{2} = 4.5$, $\frac{4 \cdot 1}{2} = 2$. What remains is $16 - 8.5 = 7.5$.
- PXZ: enclosing rectangle $6 \times 3 = 18$; subtract $\frac{4 \cdot 3}{2} = 6$, $\frac{2 \cdot 2}{2} = 2$, $\frac{6 \cdot 1}{2} = 3$. What remains is $18 - 11 = 7$.

- PVX: enclosing rectangle $4 \times 4 = 16$; subtract $\frac{1 \cdot 4}{2} = 2$, $\frac{3 \cdot 1}{2} = 1.5$, $\frac{4 \cdot 3}{2} = 6$. What remains is $16 - 9.5 = 6.5$.



Step 3. Compare

$7.5 > 7 > 6.5 > 6 = 6$. The largest triangle is PVU.

Common mistake. Do not guess by eye: triangle PXZ looks "big" because it stretches all the way to side RS, but its area is only 7, less than that of PVU.

Check: If the side of each square is 1, all areas are divided by 4 (1.875; 1.75; 1.625; 1.5; 1.5), and the order does not change.

Option **(A)**.

Answer: (A) triangle PVU

Extension and challenge

Problem 31

In the coordinate plane Oxy, consider the point $A(-3,\ 2)$.

a) Let B be the reflection of A in the x-axis, C the reflection of A in the y-axis, and D the reflection of A in the origin O. Find the coordinates of B, C, D and state the quadrant of each point.

b) The four points A, C, D, B are the vertices of a rectangle. Find the perimeter and the area of that rectangle.

Detailed solution

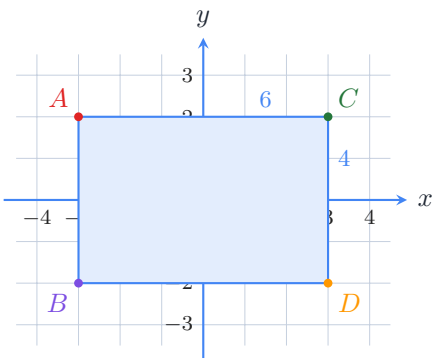
Understanding the problem. "Reflection in an axis" is like looking in a mirror placed along that axis: the image lies on the other side of the axis, at the same distance from it as the original point. We check which coordinate changes sign and which stays the same.

> **Recall.** Reflection in the x-axis: $(x,\ y) \to (x,\ -y)$. Reflection in the y-axis: $(x,\ y) \to (-x,\ y)$. Reflection in the origin O (also a 180° rotation about O): $(x,\ y) \to (-x,\ -y)$.

Step 1. Point B – reflection in Ox
Mirroring in the x-axis: the point keeps its left–right position (the x-coordinate is unchanged) and only moves from above to below (the y-coordinate changes sign). $B(-3,\ -2)$, of the form $(-, -)$: quadrant III.

Step 2. Point C – reflection in Oy
Mirroring in the y-axis: the y-coordinate is unchanged, the x-coordinate changes sign. $C(3,\ 2)$, of the form $(+, +)$: quadrant I.

Step 3. Point D – reflection in the origin O
Both coordinates change sign: $D(3,\ -2)$, of the form $(+, -)$: quadrant IV.



Step 4. Find the dimensions of the rectangle
$A(-3,\ 2)$ and $C(3,\ 2)$ have the same y-coordinate: $AC = 3 - (-3) = 6$. $A(-3,\ 2)$ and $B(-3,\ -2)$ have the same x-coordinate: $AB = 2 - (-2) = 4$.

Step 5. Find the perimeter and the area

Recall. Rectangle with length a, width b: $P = 2 \cdot (a + b)$, $S = a \cdot b$.

$P = 2 \cdot (6 + 4) = 20$ (units); $S = 6 \cdot 4 = 24$ (square units).

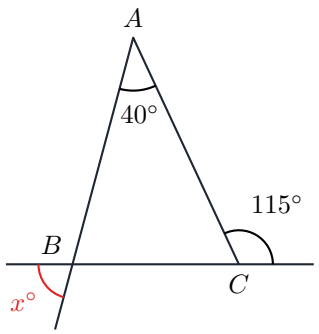
ThinkAlgo Tip. This rectangle has both coordinate axes as lines of symmetry. Each quadrant contains exactly one quarter of it, a 3×2 rectangle of area 6; and $4 \cdot 6 = 24$.

Check: Each quadrant contains one point: A (II), C (I), B (III), D (IV), as in the figure.

Answer: a) $B(-3, -2)$**: III;** $C(3, 2)$**: I;** $D(3, -2)$**: IV. b) Perimeter 20, area 24.**

Problem 32

In the figure, triangle ABC has $\widehat{BAC} = 40°$; side BC is extended in both directions and side AB is extended beyond B. The exterior angle at vertex C is 115°. The value of x is



(A) 65 (B) 75 (C) 105 (D) 115 (E) 140

Detailed solution

Understanding the problem. Angle $x°$ at vertex B is formed by two extensions (of BC and of BA), so it is *vertically opposite* to the interior angle $\widehat{ABC}$. So we only need $\widehat{ABC}$. We solve it in two ways.

Step 1. Method 1 – via the interior angle at C

Recall. Two supplementary angles add up to 180°; the three interior angles of a triangle add up to 180°.

The interior angle at C is supplementary to the exterior angle 115°: $\widehat{ACB} = 180° - 115° = 65°$. Interior angle at B: $\widehat{ABC} = 180° - 40° - 65° = 75°$.

Step 2. Method 2 – the exterior angle property

Recall. An exterior angle of a triangle equals the sum of the two interior angles not adjacent to it.

The exterior angle at C equals $\widehat{A} + \widehat{B}$: $115° = 40° + \widehat{ABC}$, so $\widehat{ABC} = 115° - 40° = 75°$.

Step 3. Deduce x

Recall. Vertically opposite angles are equal.

$x° = \widehat{ABC} = 75°$, so $x = 75$.

Common mistake.

- (A) 65 is the interior angle at C, not at B.
- (C) 105 is the angle supplementary to $\widehat{ABC}$ – confusing "supplementary" with "vertically opposite".

Check: $40° + 75° + 65° = 180°$ and $40° + 75° = 115°$ (exterior angle at C). Both conditions hold.

So the answer is **(B)**.

Answer: (B) 75

Problem 33

a) In an isosceles triangle, the apex angle is 4 times each base angle. Find the angles of the triangle.

b) An isosceles triangle has an angle of 36°. Find all possibilities for the other two angles. For each possibility, find the difference between the largest and the smallest angle of the triangle.

c) Is there an isosceles triangle with a base angle of 92°? Why?

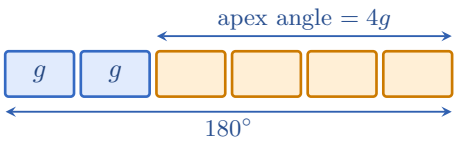
Detailed solution

Understanding the problem. All three parts rest on two facts: an isosceles triangle has two equal base angles, and the three angles add up to 180°.

Recall. Isosceles triangle: apex angle d, each base angle g, then $d + 2g = 180°$.

Step 1. Part a – using "equal parts"

Let each base angle be 1 part. The apex angle is 4 times as large, so it is 4 parts. The three angles together make $1 + 1 + 4 = 6$ parts, corresponding to 180°.



One part: $180° \div 6 = 30°$. So each base angle is $30°$ and the apex angle is $4 \cdot 30° = 120°$.

Step 2. Part b – two cases

Case 1: $36°$ *is the apex angle.* Each base angle: $(180° - 36°) \div 2 = 144° \div 2 = 72°$. Triangle $36°$, $72°$, $72°$. Difference between largest and smallest: $72° - 36° = 36°$.

Case 2: $36°$ *is a base angle.* The other base angle is $36°$, the apex angle $180° - 72° = 108°$. Triangle $36°$, $36°$, $108°$. Difference: $108° - 36° = 72°$.

Step 3. Part c – reasoning

Why is it true? If one base angle were $92°$, the other base angle would also be $92°$. Then the two base angles alone would add up to $184°$, more than $180°$ – impossible, since *all three* angles add up to only $180°$. So no such isosceles triangle exists. (In general: a base angle of an isosceles triangle is always less than $90°$.) □

Check: $30 + 30 + 120 = 180$; $36 + 72 + 72 = 180$; $36 + 36 + 108 = 180$.

Answer: a) $30°$, $30°$, $120°$. b) $72°$ and $72°$ (difference $36°$), or $36°$ and $108°$ (difference $72°$). c) No.

Problem 34*

In the coordinate plane, consider the point $A(4, 1)$.

a) Rotating A about the origin O through $90°$ counterclockwise gives the point B; rotating B a further $90°$ counterclockwise gives C; rotating C a further $90°$ counterclockwise gives E. Find the coordinates of B, C, E.

b) Rotating A about O through $270°$ *clockwise* gives which of these three points?

c) The quadrilateral $ABCE$ is a square. Find its area.

Detailed solution

Understanding the problem. We use the rule for a $90°$ rotation in the coordinate plane, then find the area of the "tilted" square by placing it inside a big square whose sides run along the grid lines.

Recall. Rotation of $90°$ counterclockwise about O: $(x, y) \to (-y, x)$. Rotating α clockwise is the same as rotating $360° - \alpha$ counterclockwise.

Why is it true? Why $(x, y) \to (-y, x)$? The point $A(4, 1)$ is reached by going "right 4, up 1". Turn this whole path $90°$ counterclockwise: "right" becomes "up", and "up" becomes "left". So the image is reached by "up 4, left 1", i.e. the point $(-1, 4)$. □

Step 1. Part a – apply the rule three times

$A(4, 1) \to B(-1, 4) \to C(-4, -1) \to E(1, -4)$.

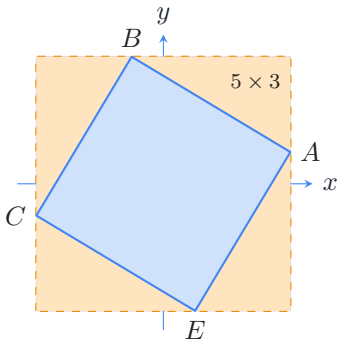
(Check: C is the image of A under a $180°$ rotation, which is indeed $(-4, -1)$, both coordinates changing sign.)

Step 2. Part b

Rotating $270°$ clockwise is the same as rotating $360° - 270° = 90°$ counterclockwise, so we get

exactly the point $B(-1, 4)$.

Step 3. Part c – put the square in a frame



Square $ABCE$ fits inside the big square with vertices $(\pm 4, \pm 4)$, of side $4 - (-4) = 8$ and area $8 \cdot 8 = 64$.

The extra parts are 4 equal right triangles in the four corners. For example, the triangle in the top-right corner has vertices $A(4, 1)$, $(4, 4)$, $B(-1, 4)$: its horizontal leg is $4 - (-1) = 5$, its vertical leg is $4 - 1 = 3$, and its area is $\dfrac{5 \cdot 3}{2} = 7.5$.

Area of square $ABCE$: $64 - 4 \cdot 7.5 = 64 - 30 = 34$ (square units).

ThinkAlgo Tip. "Frame it and subtract" is a very powerful way to find the area of a polygon whose vertices are grid points: draw the smallest surrounding rectangle, then subtract the right triangles (trapezoids, rectangles) outside the polygon.

Check: The four equal right triangles (each 5×3) together with square $ABCE$ fill the 8×8 frame: $4 \cdot 7.5 + 34 = 64$. Also, the frame side $8 = 5 + 3$ is exactly the sum of the two legs.

Answer: a) $B(-1, 4)$, $C(-4, -1)$, $E(1, -4)$. **b) Point** B. **c)** 34 **square units.**

Problem 35

Fill in the blanks:

a) 2.4 km = ... m; b) 75 mm = ... cm; c) 3 m^2 = ... cm^2; d) 2500 cm^3 = ... litres.

e) A rectangular board is 1.2 m long and 50 cm wide. Find the area of the board in cm^2 and in m^2.

Detailed solution

Understanding the problem. Each part is a unit conversion. Key point: lengths go up by a factor of 10 per step, areas by 100, volumes by 1000.

Recall. 1 km = 1000 m; 1 cm = 10 mm; 1 m^2 = 10 000 cm^2; 1 litre = 1 dm^3 = 1000 cm^3. Larger to smaller: multiply; smaller to larger: divide.

Step 1. Part a
km is larger than m, so multiply: $2.4 \cdot 1000 = 2400$. So 2.4 km = 2400 m.

Step 2. Part b
mm is smaller than cm, so divide: $75 \div 10 = 7.5$. So 75 mm = 7.5 cm.

Step 3. Part c
$3\ \text{m}^2 = 3 \cdot 10\,000\ \text{cm}^2 = 30\,000\ \text{cm}^2$.

Common mistake. Converting an area by multiplying only by 100 (as for lengths in m to cm) gives 300 – wrong. A 1 m × 1 m square is 100 cm × 100 cm and contains 10 000 squares of 1 cm^2.

Step 4. Part d
$2500\ \text{cm}^3 = 2500 \div 1000\ \text{dm}^3 = 2.5\ \text{dm}^3 = 2.5$ litres.

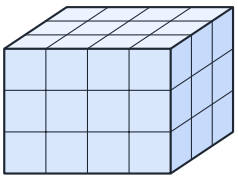
Step 5. Part e – use the same unit first
The two dimensions are in different units, so convert before multiplying: 1.2 m = 120 cm. Area: $120 \cdot 50 = 6000$ (cm^2). In m^2: $6000 \div 10\,000 = 0.6$ (m^2).

Check: Directly in metres: 50 cm = 0.5 m, so $1.2 \cdot 0.5 = 0.6$ (m^2). It matches.

Answer: a) 2400; b) 7.5; c) 30 000; d) 2.5; e) 6000 $\text{cm}^2 = 0.6\ \text{m}^2$.

Problem 36

An built a 4 cm × 3 cm × 3 cm box from 36 small cubes of edge 1 cm (as in Problem 10). An paints all the outside faces of the box and then takes the small cubes apart. How many small cubes have exactly 3 painted faces? Exactly 2? Exactly 1? No painted face?



Detailed solution

Understanding the problem. The number of painted faces of a small cube depends on its *position*: at a corner, along an edge, in the middle of a face, or completely inside. We count each type.

Step 1. Corner cubes – 3 painted faces
Each corner cube of the box shows 3 faces (one front/back, one top/bottom, one left/right). The box has 8 corners, so **8** cubes have 3 painted faces.

Step 2. Cubes along an edge (not at the ends) – 2 painted faces
The box has 12 edges: 4 edges of length 4 cm, 4 edges of length 3 cm (height) and 4 edges of length 3 cm (depth). Along an edge of n cubes, remove the two end cubes (already counted in step 1): $n - 2$ cubes remain.

- 4 edges of length 4: each $4 - 2 = 2$ cubes, total $4 \cdot 2 = 8$;
- 8 edges of length 3: each $3 - 2 = 1$ cube, total $8 \cdot 1 = 8$.

So $8 + 8 = \mathbf{16}$ cubes have 2 painted faces.

Step 3. Cubes in the middle of a face – 1 painted face

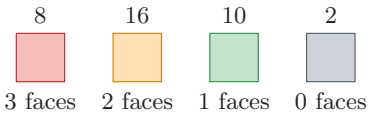
On an $m \times n$ face, removing the outer border leaves $(m-2)\cdot(n-2)$ cubes.

- 2 faces 4×3 (front, back): each $2 \cdot 1 = 2$ cubes, total 4;
- 2 faces 4×3 (top, bottom): each $2 \cdot 1 = 2$ cubes, total 4;
- 2 faces 3×3 (left, right): each $1 \cdot 1 = 1$ cube, total 2.

So $4 + 4 + 2 = \mathbf{10}$ cubes have 1 painted face.

Step 4. Cubes completely inside – unpainted

Removing one layer of "skin" on each side, the inner core measures $(4-2)\times(3-2)\times(3-2) = 2\times1\times1$, i.e. **2** cubes.



Note for parents. If you have a set of toy cubes (or a Rubik's-type cube), build the $4 \times 3 \times 3$ box with your child, stick coloured paper on the outside faces, then take it apart and count. Your child will see for themselves why a corner cube has 3 coloured faces.

Check: The four types must add up to the number of cubes: $8 + 16 + 10 + 2 = 36 = 4 \cdot 3 \cdot 3$. Correct.

Answer: 3 faces: 8 cubes; 2 faces: 16 cubes; 1 face: 10 cubes; unpainted: 2 cubes.

Problem 37

A rectangle has the same perimeter as a square of area 81 cm^2. The length of the rectangle is twice its width. The area of the rectangle is

(A) 81 cm^2 (B) 72 cm^2 (C) 54 cm^2 (D) 36 cm^2 (E) 64 cm^2

Detailed solution

Understanding the problem. The chain of reasoning: area of the square → side of the square → perimeter (shared by both figures) → half-perimeter of the rectangle → length and width → area.

Recall. Square with side a: $S = a \cdot a$, $P = 4a$. Rectangle: $P = 2 \cdot (\text{length} + \text{width})$, $S = \text{length} \cdot \text{width}$.

Step 1. Side of the square

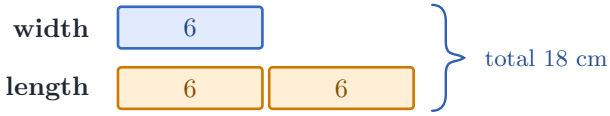
We need a number a with $a \cdot a = 81$: $a = 9$ cm (since $9 \cdot 9 = 81$).

Step 2. Perimeter

Perimeter of the square $= 4 \cdot 9 = 36$ cm, which is also the perimeter of the rectangle.

Step 3. Sum of length and width
Half-perimeter of the rectangle: $36 \div 2 = 18$ cm = length + width.

Step 4. Split in the given ratio
The length is twice the width: width is 1 part, length is 2 parts, 3 parts in all corresponding to 18 cm.



One part $= 18 \div 3 = 6$ cm. Width 6 cm, length 12 cm.

Step 5. Area
$S = 12 \cdot 6 = 72$ (cm^2).

ThinkAlgo Tip. Both figures have perimeter 36 cm, but the square's area 81 cm^2 is larger than the rectangle's (72 cm^2). Among all rectangles with the same perimeter, the square has the largest area.

Check: Perimeter of the rectangle: $2 \cdot (12 + 6) = 36$ cm = perimeter of the square; $12 = 2 \cdot 6$. Correct.
So the answer is **(B)**.

Answer: (B) 72 cm^2

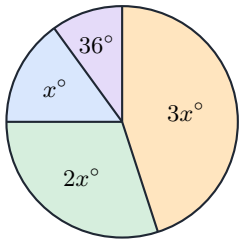
Problem 38 *

A circular spinner is divided into four regions with central angles $36°$, $x°$, $2x°$ and $3x°$ (see figure).

a) Find x.

b) What percentage of the spinner's area does each region take up?

c) About what percentage of spins will *not* stop in the largest region?



Detailed solution

Understanding the problem. The four central angles together make a full turn. This gives an equation for x; then we convert each angle to a percentage.

Recall. The angles around a point add up to 360°. A region with central angle $n°$ is $\frac{n}{360}$ of the circle.

Step 1. Part a – set up the equation
$36 + x + 2x + 3x = 360$, i.e. $36 + 6x = 360$. Subtract 36 from both sides: $6x = 324$. Divide by 6: $x = 54$.

Step 2. Part b – the angles and percentages
The four angles: 36°, 54°, 108°, 162°.

$$\frac{36}{360} = 10\%, \qquad \frac{54}{360} = 15\%, \qquad \frac{108}{360} = 30\%, \qquad \frac{162}{360} = 45\%.$$

(Shortcut: $\frac{36}{360} = \frac{1}{10}$; the other angles are 1.5, 3 and 4.5 times 36°, so the percentages are 15%, 30%, 45%.)

Step 3. Part c – use the complement
The largest region is the 162° one, with 45%. The pointer does *not* stop there in about $100\% - 45\% = 55\%$ of spins.

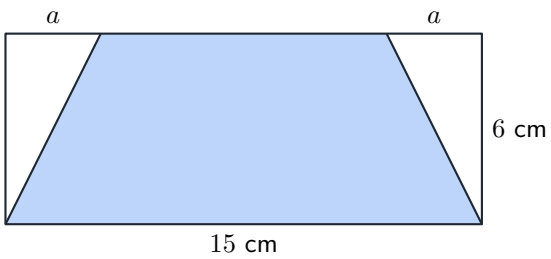
Common mistake. The "55%" is only *approximately* the share of spins after a great many spins. In just a few spins the actual count may differ; the area ratio tells how *often* the pointer stops somewhere in the long run.

Check: $36 + 54 + 108 + 162 = 360$ and $10\% + 15\% + 30\% + 45\% = 100\%$.

Answer: a) $x = 54$. **b)** 10%, 15%, 30%, 45%. **c) About** 55%.

Problem 39

A rectangle is 15 cm long and 6 cm wide. On its top side, two segments of the same length a cm are taken at the two ends, and the two right triangles at the top corners are cut off, leaving the shaded trapezoid shown. The area of the trapezoid is 72 cm^2. Find a and the length of the shorter parallel side of the trapezoid.



Detailed solution

Understanding the problem. Problem 14 gave the dimensions and asked for the area; this problem goes the other way: the area is given and a dimension is asked for. We express the trapezoid's area in terms of a as "rectangle minus two triangles", then find a.

Recall. $S_{\text{rectangle}} = \text{length} \times \text{width}$; $S_{\text{right triangle}} = \dfrac{\text{product of the two legs}}{2}$.

Step 1. Area of the rectangle
$15 \cdot 6 = 90$ (cm^2).

Step 2. Total area of the two cut-off triangles
Area of the two triangles $= 90 - 72 = 18$ (cm^2), so each triangle has area $18 \div 2 = 9$ (cm^2).

Step 3. Find a
Each right triangle has legs a and 6:

$$\frac{a \cdot 6}{2} = 9 \quad \Rightarrow \quad 3 \cdot a = 9 \quad \Rightarrow \quad a = 3.$$

Step 4. Shorter parallel side of the trapezoid
Shorter side $= 15 - a - a = 15 - 3 - 3 = 9$ cm.

Check: By the trapezoid formula: $\dfrac{(15+9)\cdot 6}{2} = \dfrac{24 \cdot 6}{2} = 72$ (cm^2). Correct.

ThinkAlgo Tip. It can also be solved with the trapezoid formula: the shorter side b satisfies $\dfrac{(15+b)\cdot 6}{2} = 72$, i.e. $(15+b)\cdot 3 = 72$, $15+b = 24$, $b = 9$; then $a = (15-9) \div 2 = 3$.

Answer: $a = 3$ cm; shorter parallel side 9 cm.

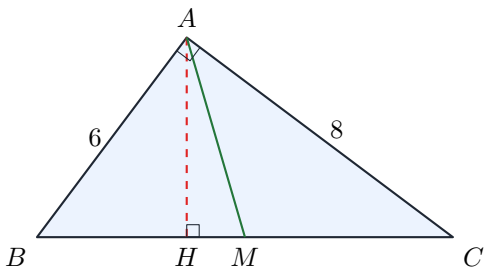
Problem 40 *

Triangle ABC has a right angle at A, with $AB = 6$ cm, $AC = 8$ cm and $BC = 10$ cm.

a) Find the area of triangle ABC.

b) Find the length of the height AH drawn from A to side BC.

c) Let M be the midpoint of BC. Find the area of triangle ABM.



Detailed solution

Understanding the problem. Any side of a triangle may be chosen as the base, as long as we use the height belonging to that base. The area is always the same. We compute the area the easy way (base AB, height AC), then use it to find the height to the base BC.

Recall. $S = \dfrac{\text{base} \cdot \text{height}}{2}$, so height $= \dfrac{2 \cdot S}{\text{base}}$. In a right triangle, each leg is the height to the other leg.

Step 1. Part a – choose base AB

Since angle A is a right angle, AC is perpendicular to AB: with AB as the base, AC is the height.

$$S_{ABC} = \frac{6 \cdot 8}{2} = 24 \text{ (cm}^2\text{)}.$$

Step 2. Part b – choose base BC

With $BC = 10$ cm as the base, the matching height is AH. The area is still 24 cm^2:

$$\frac{10 \cdot AH}{2} = 24 \quad \Rightarrow \quad 10 \cdot AH = 48 \quad \Rightarrow \quad AH = 4.8 \text{ cm}.$$

Step 3. Part c – the midpoint halves the area

Recall. Two triangles with equal bases and a common height have equal areas.

Triangles ABM and ACM have bases $BM = MC = 5$ cm (since M is the midpoint) and the common height AH (both drawn from A to the line BC). So they have equal areas, each half of triangle ABC:

$$S_{ABM} = \frac{5 \cdot 4.8}{2} = 12 \text{ (cm}^2\text{)} = \frac{24}{2}.$$

Common mistake. Do not use "base BC times side AB" – side AB is not perpendicular to BC, so it is not the height to the base BC.

Check: $AH = 4.8$ is smaller than both $AB = 6$ and $AC = 8$ – sensible, because the perpendicular is the shortest path from A to the line BC. And $\frac{10 \cdot 4.8}{2} = 24$.

Answer: a) 24 cm^2. b) $AH = 4.8$ cm. c) 12 cm^2.

Problem 41 *

(Extension of Problem 16.) We only consider rectangles whose two side lengths are *positive integers* (in cm); a square also counts as a rectangle.

a) Find all rectangles whose area (in cm^2) is numerically *equal* to their perimeter (in cm).

b) Find all rectangles whose area is numerically *twice* their perimeter.

Detailed solution

Understanding the problem. Call the sides a and b (positive integers). Without loss of generality, assume $a \le b$ (a is the shorter side). The area is ab and the perimeter is $2(a+b)$. This is a "whole-number solutions" problem: one equation with two unknowns. Trick: use $a \le b$ to *trap* a in a small range, then try each value.

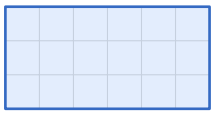
a) Condition: $ab = 2a + 2b$.

Step 1. An upper bound for a
Since $a \le b$, we have $2a + 2b \le 2b + 2b = 4b$. Hence $ab \le 4b$. Dividing both sides by $b > 0$: $a \le 4$.

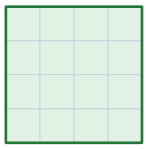
Step 2. A lower bound for a
Since $2a > 0$, we have $ab = 2a + 2b > 2b$. Dividing both sides by $b > 0$: $a > 2$. So $a = 3$ or $a = 4$.

Step 3. Try each value

- $a = 3$: $3b = 6 + 2b \Rightarrow b = 6$. The 3×6 rectangle.
- $a = 4$: $4b = 8 + 2b \Rightarrow 2b = 8 \Rightarrow b = 4$. The 4×4 square.



$A = 18,\ P = 18$



$A = 16,\ P = 16$

b) Condition: $ab = 2 \cdot 2(a + b) = 4a + 4b$.

Step 4. Trap a as before
$ab = 4a + 4b \le 8b \Rightarrow a \le 8$; and $ab = 4a + 4b > 4b \Rightarrow a > 4$. So $a \in \{5, 6, 7, 8\}$.

Step 5. Try each value
From $ab = 4a + 4b$ we get $ab - 4b = 4a$, i.e. $b(a - 4) = 4a$, so $b = \dfrac{4a}{a - 4}$.

a	5	6	7	8
$b = \dfrac{4a}{a-4}$	$\dfrac{20}{1} = 20$	$\dfrac{24}{2} = 12$	$\dfrac{28}{3}$ (not an integer)	$\dfrac{32}{4} = 8$

There are three rectangles: 5×20, 6×12, 8×8.

Why is it true? Why have we missed none? Every rectangle that works has its shorter side a in the range we trapped, and for each a the side b is determined uniquely by the equation. We tried all those values of a. □

ThinkAlgo Tip. Another way for part a): $ab - 2a - 2b + 4 = 4$, i.e. $(a - 2)(b - 2) = 4 = 1 \cdot 4 = 2 \cdot 2$. Hence $a - 2 = 1$, $b - 2 = 4$ or $a - 2 = b - 2 = 2$. Again we get 3×6 and 4×4.

Check: 3×6: $A = 18$, $P = 2 \cdot 9 = 18$. 4×4: $16 = 16$. 5×20: $A = 100$, $P = 50$; 6×12: $A = 72$, $P = 36$; 8×8: $A = 64$, $P = 32$ – all twice.

Answer: a) 3×6 and 4×4. b) 5×20, 6×12, 8×8.

Problem 42 *

(Extension of Problem 17.) The letter P is written in the top-right cell of a 2×2 grid: 

. We may use three transformations, as many times as we like and in any order:

- Q: rotate the grid 90° clockwise about its centre;
- D: reflect the grid in the *vertical* line through the centre;
- N: reflect the grid in the *horizontal* line through the centre.

a) Draw the result of doing D and then N. Can that result be obtained using only Q?

b) Show that doing D and then Q has the same effect as a reflection in one of the *diagonals* of the grid. Which diagonal?

c) How many different pictures can be obtained altogether (including the starting picture)? Explain why there cannot be more.

Detailed solution

Understanding the problem. As in Problem 17, think of the grid as a clear square *card* with the letter P drawn on it. Each transformation lifts the card and puts it back exactly in its place. To keep track we record two things: which cell P is in, and how it has been turned or flipped.

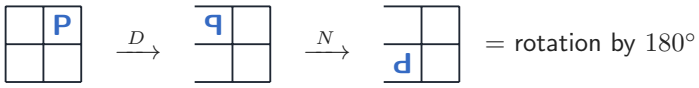
a)

Step 1. Do D

Reflecting in the vertical line: the top-right cell goes to the top-left cell, and P is "mirrored" (its bowl now faces left).

Step 2. Then do N

Reflecting in the horizontal line: the top-left cell goes down to the bottom-left cell, and the letter is flipped top-to-bottom. Result: P is in the bottom-left cell, *upside down* (looking like a "d"), as in the third picture below.



Step 3. Compare with the Q moves

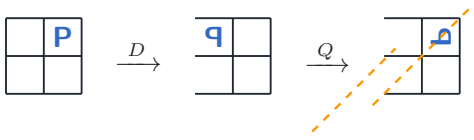
Doing Q twice (a 180° rotation): the top-right cell goes to the opposite cell through the centre, the bottom-left cell, and the letter is upside down. This is the same result. So **D then N has the same effect as Q then Q.**

> **Recall.** Two reflections in two perpendicular lines (both through the centre) have the same effect as a 180° rotation about the centre.

b)

Step 4. Do D and then Q

D: P goes to the top-left cell, mirrored. Q (90° clockwise): the top-left cell goes to the top-right cell. Result: P is *back in the top-right cell*, but its stem is horizontal and its bowl is on top:



Step 5. Recognise the diagonal reflection

The top-right cell has not moved, which means it lies on the mirror line. The diagonal through the top-right cell (and the bottom-left cell) is the diagonal joining the **bottom-left corner to the top-right corner** (dashed). Reflecting in this line turns "up" into "right" and vice versa, so the (vertical) stem of P becomes horizontal and the bowl of P (on the right) moves to the top – exactly as in the picture. A further check: a letter in the top-left cell moves to the bottom-right cell in both cases.

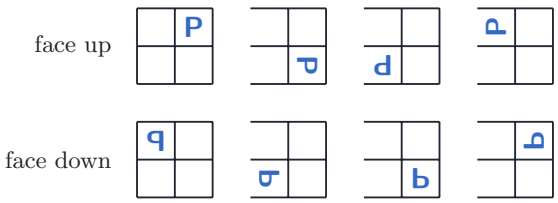
c)

Step 6. There cannot be more than 8 pictures

Why is it true? Mark the top-right corner of the card with a star. Each time the card is put back, the star can only be in 1 of the 4 corners of the frame. In addition, the card can only be *face up* or *face down* (each reflection D or N switches face up and face down; the rotation Q does not). Once we know where the star is and whether the card is face up or face down, the position of the card – and hence the whole picture – is completely determined. So there are at most $4 \times 2 = 8$ pictures. □

Step 7. All 8 pictures occur

Four "face up" pictures: rotations by 0°, 90°, 180°, 270° (using Q from 0 to 3 times). Four "face down" pictures: each of the above followed by D. The letter P has no line of symmetry, so these 8 pictures are all different:

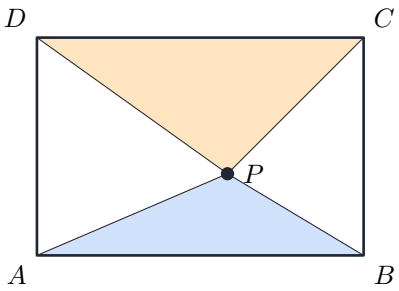


Common mistake. Do not count "4 cells × 8 orientations = 32". The cell and the orientation of the letter are *tied together*, because both move with the same card.

Answer: a) P in the bottom-left cell, upside down; the same as Q then Q (a 180° rotation). b) The diagonal joining the bottom-left corner to the top-right corner. c) 8 pictures.

Problem 43

(Extension of Problem 18.) Rectangle $ABCD$ has $AB = 12$ cm and $BC = 8$ cm. Point P lies inside the rectangle. (We write S_{XYZ} for the area of triangle XYZ.)



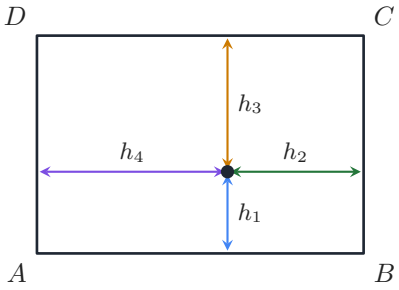
a) Prove that wherever P is, we always have $S_{PAB} + S_{PCD} = 48$ cm^2.

b) Given $S_{PAB} = 18$ cm^2 and $S_{PBC} = 20$ cm^2, find S_{PCD}, S_{PDA} and the distances from P to the four sides of the rectangle.

c) Is there a point P inside the rectangle with $S_{PAB} = 20$, $S_{PBC} = 30$ and $S_{PCD} = 30$ (cm^2)?

Detailed solution

Understanding the problem. Each of the four triangles has a side of the rectangle as its base and P as its apex. The height of each triangle is just *the distance from P to that side*. Call the distances from P to AB, BC, CD, DA respectively h_1, h_2, h_3, h_4.



a)

Step 1. The sum of two heights

Draw through P the line perpendicular to AB; since $AB \parallel CD$ it is also perpendicular to CD. The part between AB and CD has length $BC = 8$, and P divides it into two parts h_1 and h_3. So $h_1 + h_3 = 8$. Similarly $h_2 + h_4 = AB = 12$.

Step 2. Add the two areas

> **Recall.** Area of a triangle $= \frac{1}{2} \times$ base $\times$ height.

$$S_{PAB} + S_{PCD} = \frac{12 \cdot h_1}{2} + \frac{12 \cdot h_3}{2} = 6(h_1 + h_3) = 6 \cdot 8 = 48 \text{ (cm}^2\text{)}.$$

The result does not depend on the position of P. (It is exactly half the area of the rectangle, $12 \cdot 8 = 96$.) Similarly, $S_{PBC} + S_{PDA} = \frac{8}{2}(h_2 + h_4) = 4 \cdot 12 = 48$ cm^2.

b)

Step 3. **Use part a**
$S_{PCD} = 48 - 18 = 30$ cm^2, $S_{PDA} = 48 - 20 = 28$ cm^2.

Step 4. **Compute the four distances**
From $A = \frac{1}{2} \cdot$ base $\cdot h$ we get $h = 2A \div$ base.

$$h_1 = \frac{2 \cdot 18}{12} = 3, \quad h_2 = \frac{2 \cdot 20}{8} = 5, \quad h_3 = \frac{2 \cdot 30}{12} = 5, \quad h_4 = \frac{2 \cdot 28}{8} = 7 \text{ (cm)}.$$

c)

Step 5. **Check the compulsory condition**
By part a), we always have $S_{PAB} + S_{PCD} = 48$. But $20 + 30 = 50 \neq 48$. So **there is no** such point P.

ThinkAlgo Tip. Problem 18 is a special case: P is the centre of a square, so the four triangles are equal. Here P is arbitrary, yet "the sum of two opposite triangles" is still always half of the figure.

Check: $h_1 + h_3 = 3 + 5 = 8 = BC$ and $h_2 + h_4 = 5 + 7 = 12 = AB$. With $A(0, 0)$ we get $P(7, 3)$ – exactly the point in the picture.

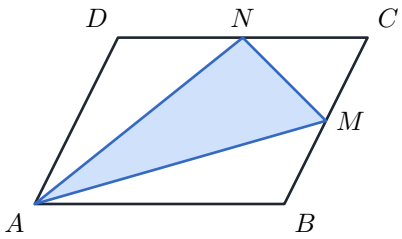
Answer: b) $S_{PCD} = 30$ **cm^2,** $S_{PDA} = 28$ **cm^2; the distances to** AB, BC, CD, DA **are** $3, 5, 5, 7$ **cm. c) No.**

Problem 44

(Extension of Problem 22.) Parallelogram $ABCD$ has area 48 cm^2.

a) Let M and N be the midpoints of BC and CD respectively. Find the area of triangle AMN.

b) Keep N as the midpoint of CD, but take M on side BC with $BM = 3\,MC$. Find the area of triangle AMN.



Detailed solution

Understanding the problem. Triangle AMN sits "diagonally" inside the parallelogram. We take the whole parallelogram and remove the three corner triangles ABM, MCN and NDA (as usual, S_{XYZ} denotes the area of triangle XYZ):

$$S_{AMN} = S_{ABCD} - S_{ABM} - S_{MCN} - S_{NDA}.$$

Recall. (1) Each diagonal divides a parallelogram into two triangles of equal area ($= 24$ cm^2). (2) Two triangles with a common apex whose bases lie on the same line have areas in the same ratio as their bases.

a)

Step 1. Triangle ABM
Compared with triangle ABC ($= 24$): common apex A, base $BM = \frac{1}{2}BC$. So $S_{ABM} = \frac{1}{2} \cdot 24 = 12$.

Step 2. Triangle NDA
Compared with triangle ACD ($= 24$): common apex A, base $DN = \frac{1}{2}DC$. So $S_{NDA} = 12$.

Step 3. Triangle MCN (in two steps, as in Problem 22)
Triangle BCD has area 24. Triangle MCD shares the apex D with it, with base $CM = \frac{1}{2}CB$: $S_{MCD} = 12$. Triangle MCN shares the apex M with triangle MCD, with base $CN = \frac{1}{2}CD$: $S_{MCN} = \frac{1}{2} \cdot 12 = 6$.

Step 4. Result
$S_{AMN} = 48 - 12 - 6 - 12 = 18$ cm^2.

b)

Step 5. Redo it with $MC = \frac{1}{4}BC$
Since $BM = 3\,MC$, we have $BC = 4\,MC$, i.e. $MC = \frac{1}{4}BC$ and $BM = \frac{3}{4}BC$.

- $S_{ABM} = \frac{3}{4} \cdot 24 = 18$;
- $S_{NDA} = 12$ (unchanged);
- $S_{MCD} = \frac{1}{4} \cdot 24 = 6$, then $S_{MCN} = \frac{1}{2} \cdot 6 = 3$.

$S_{AMN} = 48 - 18 - 3 - 12 = 15$ cm^2.

ThinkAlgo Tip. The "corner" rule: if on two sides of a triangle, measured from the same vertex, we take the parts $\frac{p}{q}$ and $\frac{r}{s}$ of those sides, then the small triangle at that corner has area $\frac{p}{q} \cdot \frac{r}{s}$ of the big triangle. In part b): $\frac{1}{4} \cdot \frac{1}{2} = \frac{1}{8}$ of 24 is 3.

Check: Put $A(0,0)$, $B(8,0)$, $D(2,6)$, $C(10,6)$ (area $8 \cdot 6 = 48$). Part a): $M(9,3)$, $N(6,6)$; enclose triangle AMN in a $9 \times 6 = 54$ rectangle and subtract $\frac{9 \cdot 3}{2} = 13.5$, $\frac{3 \cdot 3}{2} = 4.5$, $\frac{6 \cdot 6}{2} = 18$: 18 remains. Correct.

Answer: a) 18 cm^2. b) 15 cm^2.

Problem 45 *

(Extension of Problem 21.) Consider triangles whose three side lengths are positive integers and whose perimeter is 15 (two triangles with the same three side lengths count as one).

a) Let c be the longest side. Prove that $5 \le c \le 7$.

b) How many such triangles are there? List them.

c) How many of them are isosceles? How many have *exactly* two equal sides?

Detailed solution

Understanding the problem. Call the sides $a \le b \le c$ (in increasing order, so that nothing is counted twice). We have $a+b+c=15$, and by the triangle inequality we only need to check $a+b>c$.

Recall. Three segments $a \le b \le c$ are the sides of a triangle exactly when $a+b>c$ (the two smaller sides add up to more than the largest one).

a)

Step 1. Upper bound

$a+b=15-c$. The condition $a+b>c$ becomes $15-c>c$, i.e. $2c<15$, $c<7.5$. Since c is an integer, $c \le 7$.

Step 2. Lower bound

c is the largest side, so $a \le c$ and $b \le c$, hence $15=a+b+c \le 3c$, i.e. $c \ge 5$.

b)

Step 3. List them by c

For each c we have $a+b=15-c$ and $a \le b \le c$.

c	$a+b$	Pairs (a,b) with $a \le b \le c$	Number of triangles
7	8	$(1,7)$, $(2,6)$, $(3,5)$, $(4,4)$	4
6	9	$(3,6)$, $(4,5)$ (not $(2,7)$ since $7>6$)	2
5	10	$(5,5)$	1

Altogether $4+2+1=7$ triangles:

$$(1,7,7),\ (2,6,7),\ (3,5,7),\ (4,4,7),\ (3,6,6),\ (4,5,6),\ (5,5,5).$$

Common mistake. For $c=7$, the pair $(1,7)$ gives the triangle $1,7,7$ – very "thin" but still valid because $1+7=8>7$. On the other hand, no triangle has a side 8, because the other two sides would add up to $7<8$.

c)

Step 4. Pick out the isosceles triangles

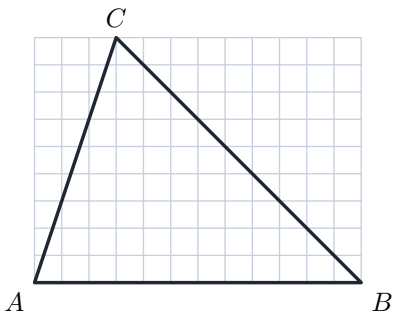
An isosceles triangle has at least two equal sides: $(1,7,7)$, $(4,4,7)$, $(3,6,6)$, $(5,5,5)$ – that is 4 triangles. Among them $(5,5,5)$ is equilateral (three equal sides), so 3 triangles have exactly two equal sides.

Check: Each triple adds up to 15 and satisfies $a+b>c$: $8>7$, $8>7$, $8>7$, $8>7$, $9>6$, $9>6$, $10>5$. Correct.

Answer: a) $5 \le c \le 7$. b) 7 triangles. c) 4 isosceles triangles, 3 of which have exactly two equal sides.

Problem 46 ★★

(Extension of Problem 23.) On a square grid (each square has side 1), triangle ABC has vertices $A(0,0)$, $B(12,0)$, $C(3,9)$. Exactly 43 points where gridlines cross lie *inside* the triangle (not on its sides). For each such point P we get three triangles PAB, PBC, PCA.



a) Find the area of triangle ABC.

b) Find the point P for which the three triangles PAB, PBC, PCA have equal areas.

c) Of all the $3 \cdot 43 = 129$ triangles, how many have an area of exactly $\frac{1}{3}$ of the area of triangle ABC? For how many positions of P is at least one of the three triangles like that?

Detailed solution

Understanding the problem. This is the "one-third version" of Problem 23. The key idea is the same: for a triangle with a fixed base, the area depends only on the height, and points with the same height lie on a line parallel to the base.

a)

Step 1. Area of triangle ABC

The base $AB = 12$ lies on the x-axis and the height is the y-coordinate of C, which is 9. Writing S_{XYZ} for the area of triangle XYZ: $S_{ABC} = \frac{12 \cdot 9}{2} = 54$. One third is 18.

c) (done first; part b) follows immediately)

Step 2. Triangle PAB with area 18

The base is $AB = 12$, so the height must be $2 \cdot 18 \div 12 = 3$, i.e. P lies on the horizontal line $y = 3$. On this line: side AC (going "1 right, 3 up") passes through $(1, 3)$; side BC (going "1 left, 1 up" from B) passes through $(9, 3)$. The points inside: $x = 2, 3, \ldots, 8$, i.e. 7 points.

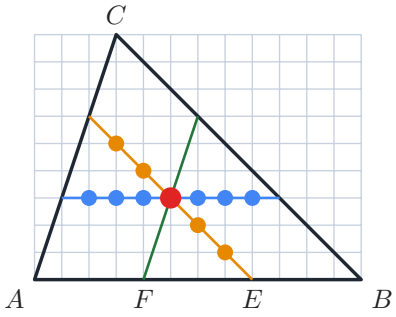
Step 3. Triangle PBC with area 18

Why is it true? Take E on AB with $EB = \frac{1}{3}AB = 4$, i.e. $E(8, 0)$. Triangles EBC and ABC share the apex C, and the base $EB = \frac{1}{3}AB$, so $S_{EBC} = \frac{1}{3} \cdot 54 = 18$. Hence $S_{PBC} = 18$ exactly when P is as far from BC as E is, i.e. when P lies on the line through E parallel to BC (on the inside of the triangle). □

BC goes in the direction "1 left, 1 up". Starting from $E(8, 0)$ in that direction: $(7, 1)$, $(6, 2)$, $(5, 3)$, $(4, 4)$, $(3, 5)$, then $(2, 6)$ – this point is on AC (on AC the y-coordinate is 3 times the x-coordinate), so it does not count. That gives 5 points.

Step 4. Triangle PCA with area 18

Similarly, take F on AB with $AF = 4$, i.e. $F(4, 0)$; P must lie on the line through F parallel to AC (direction "1 right, 3 up"): $(5, 3)$, then $(6, 6)$ – which is on BC (since $6 + 6 = 12$), so it does not count. That gives 1 point.



Step 5. **Count**

Number of triangles: $7 + 5 + 1 = 13$. The point $(5, 3)$ lies on all three lines (the red point), so it was counted three times. Number of different positions of P: $13 - 2 = 11$.

b)

Step 6. **Three equal triangles**

The three triangles have total area 54, so they are equal when each has area 18, i.e. P must lie on *all three* lines we found. There is only one such point: $P(5, 3)$.

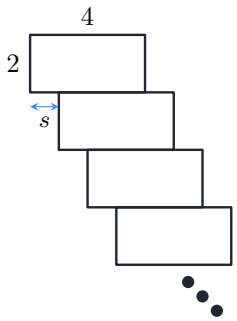
ThinkAlgo Tip. The point $(5, 3)$ is the *centroid* of the triangle: $x = \frac{0+12+3}{3} = 5$, $y = \frac{0+0+9}{3} = 3$. You will study the centroid in detail in the chapter on triangles.

Check: For $P(5, 3)$: $S_{PAB} = \frac{12 \cdot 3}{2} = 18$. S_{PCA}: enclosing rectangle $5 \times 9 = 45$, minus $\frac{3 \cdot 9}{2} = 13.5$ (top-left corner), $\frac{2 \cdot 6}{2} = 6$ (top-right corner), $\frac{5 \cdot 3}{2} = 7.5$ (bottom-right corner): 18 remains. So $S_{PBC} = 54 - 18 - 18 = 18$. Correct.

Answer: a) 54. **b)** $P(5, 3)$. **c)** 13 **triangles, at** 11 **positions of** P.

Problem 47

(Extension of Problem 25.) Again we arrange 4×2 rectangles in a "staircase" as in Problem 25, but this time the top-left vertex of each rectangle (from the second one on) is placed on the bottom side of the rectangle just above it, at distance s from that rectangle's bottom-left vertex, where s is 1, 2 or 3. (Problem 25 is the case $s = 2$.)



a) For $s = 1$ and 10 rectangles, find the perimeter of the whole figure.

b) Find a formula for the perimeter in terms of the number of rectangles n and the distance s.

c) Can we choose $s \in \{1, 2, 3\}$ and a number of rectangles n so that the perimeter is exactly 100? Find all the ways.

Detailed solution

Understanding the problem. In Problem 25 we saw that the perimeter of a "staircase" equals the perimeter of the smallest rectangle enclosing it. That argument works for every s (the top-right boundary only goes right and down; so does the bottom-left boundary). So we only need the width and the height of the whole figure.

Recall. A "staircase" shape (whose boundary on each side only goes right/down) has the same perimeter as its enclosing rectangle: $2 \times (\text{width} + \text{height})$.

b) (done first, since part a) follows from it)

Step 1. Width and height

Each new rectangle is shifted s to the right. The first has width 4 and each later one adds s: width $= 4 + (n-1)s$. Each new rectangle is 2 lower: height $= 2n$.

Step 2. The formula

$$P = 2\big(4 + (n-1)s + 2n\big).$$

a)

Step 3. Substitute $s = 1$, $n = 10$

Width $4 + 9 \cdot 1 = 13$, height 20; $P = 2(13 + 20) = 66$.

c)

Step 4. Consider each value of s

- $s = 1$: $P = 2(4 + n - 1 + 2n) = 2(3n + 3) = 6n + 6$. $6n + 6 = 100 \Rightarrow 6n = 94$: not divisible, rejected.
- $s = 2$: $P = 2(4 + 2n - 2 + 2n) = 2(4n + 2) = 8n + 4$. $8n + 4 = 100 \Rightarrow n = 12$. Accepted.
- $s = 3$: $P = 2(4 + 3n - 3 + 2n) = 2(5n + 1) = 10n + 2$. $10n + 2 = 100 \Rightarrow 10n = 98$: rejected.

There is only one way: $s = 2$ and $n = 12$.

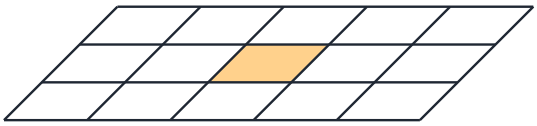
Common mistake. Remember that the height $2n$ does not depend on s; only the width changes. Many students wrongly add in the hidden pieces inside the figure.

Check: Part a) by direct counting: vertical sides $10 \cdot 4 = 40$; horizontal sides on top $4 + 9 \cdot 1 = 13$; at the bottom $9 \cdot 1 + 4 = 13$. Total 66. Correct. Part c): $s = 2$, $n = 12$: $8 \cdot 12 + 4 = 100$. Correct.

Answer: a) 66. **b)** $P = 2\big(4 + (n-1)s + 2n\big)$. **c) Only** $s = 2$, $n = 12$.

Problem 48 *

(Extension of Problem 27.) The figure below consists of 15 identical small parallelograms arranged in 3 rows of 5. The small parallelogram in the very middle is shaded.



a) How many parallelograms are there in the figure altogether?

b) How many parallelograms contain the shaded small parallelogram?

c) How many parallelograms are made of *exactly* 4 small parallelograms?

Detailed solution

Understanding the problem. The figure has 4 horizontal lines and 6 slanted lines. As in Problem 27, each parallelogram corresponds to a choice of 2 horizontal lines (top and bottom sides) and 2 slanted lines (left and right sides).

> **Recall.** The number of ways to choose 2 lines out of k lines is $\dfrac{k(k-1)}{2}$.

a)

Step 1. Count

Choose 2 of the 4 horizontal lines: $\frac{4\cdot 3}{2} = 6$ ways; choose 2 of the 6 slanted lines: $\frac{6\cdot 5}{2} = 15$ ways. Number of parallelograms: $6 \times 15 = 90$.

b)

Step 2. Condition for containing the shaded cell

Number the horizontal lines from the bottom up as $0, 1, 2, 3$; the shaded cell lies between lines 1 and 2. A parallelogram contains this cell when its bottom side is line 0 or 1 (2 ways) and its top side is line 2 or 3 (2 ways). Number the slanted lines from left to right as $0, 1, \ldots, 5$; the shaded cell lies between lines 2 and 3. The left side is line 0, 1 or 2 (3 ways); the right side is line 3, 4 or 5 (3 ways).

Step 3. Multiply

$2 \times 2 \times 3 \times 3 = 36$ parallelograms.

> **Why is it true?** Why multiply? The four choices (bottom, top, left, right) are independent: every combination gives exactly one parallelogram containing the shaded cell, and different combinations give different parallelograms. □

c)

Step 4. "Sizes" made of exactly 4 cells

A parallelogram h rows high and w columns wide consists of $h \times w$ cells. We need $h \times w = 4$ with $h \le 3$, $w \le 5$:

- $h = 1$, $w = 4$: choose the row (3 ways) and 4 adjacent columns out of 5 (2 ways): $3 \cdot 2 = 6$;
- $h = 2$, $w = 2$: choose 2 adjacent rows (2 ways) and 2 adjacent columns (4 ways): $2 \cdot 4 = 8$;
- $h = 4$, $w = 1$: impossible, since there are only 3 rows.

Total: $6 + 8 = 14$.

Common mistake. Do not forget the 2×2 type! And there are no parallelograms 4 rows high.

Check: Part a) counted by rows: 1 row high: $3 \cdot 15 = 45$; 2 rows high: $2 \cdot 15 = 30$; 3 rows high: 15 (each band has $5+4+3+2+1 = 15$ choices of columns). Total 90. It matches.

Answer: **a)** 90. **b)** 36. **c)** 14.

Problem 49 ★★

(Extension of Problems 28 and 29.) A $4 \times 5 \times 6$ rectangular box is built from 120 small cubes with edge 1. All the outside faces of the box are painted, and then the small cubes are separated.

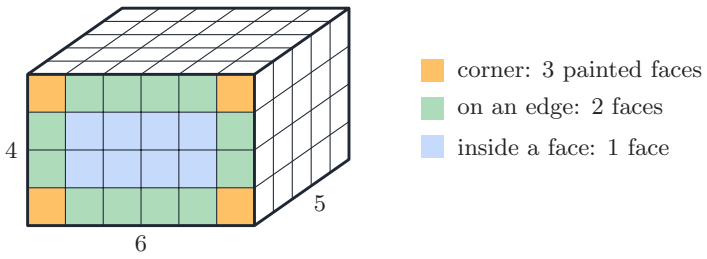
a) How many small cubes have exactly 3 painted faces? exactly 2? exactly 1? no painted face at all?

b) What is the total number of painted small faces (1×1 squares)? Explain why the result equals the surface area of the box.

c) An $n \times n \times n$ cube ($n \geq 3$) is painted and separated in the same way. Find n such that the number of small cubes with exactly 1 painted face equals the number of small cubes with no painted face.

Detailed solution

Understanding the problem. The number of painted faces of a small cube depends on its *position*: at a corner, on an edge, on a face, or completely inside. Look at the front face (size 6×4) to see this clearly:



a)

Step 1. Cubes at the corners: 3 painted faces

The box has 8 corners, each with exactly one cube: 8 cubes.

Step 2. Cubes on an edge (not at its ends): 2 painted faces

The box has 12 edges: 4 of length 4, 4 of length 5, 4 of length 6. An edge of length L contains L cubes; removing the 2 cubes at its ends (already counted as corners) leaves $L-2$ cubes:

$$4 \cdot (4-2) + 4 \cdot (5-2) + 4 \cdot (6-2) = 8 + 12 + 16 = 36.$$

Step 3. Cubes in the middle of a face: 1 painted face

On a $p \times q$ face, removing the outer border leaves $(p-2)(q-2)$ squares. The box has 2 faces 4×5, 2 faces 4×6 and 2 faces 5×6:

$$2 \cdot (2 \cdot 3) + 2 \cdot (2 \cdot 4) + 2 \cdot (3 \cdot 4) = 12 + 16 + 24 = 52.$$

Step 4. Cubes completely inside: not painted
Peel off a "shell" of thickness 1 on every side; what remains is a box of $(4-2)\times(5-2)\times(6-2) = 2\times 3\times 4 = 24$ cubes.

Check: $8+36+52+24 = 120$ – exactly the total number of cubes.

b)

Step 5. Count the painted small faces
$8\cdot 3 + 36\cdot 2 + 52\cdot 1 + 24\cdot 0 = 24+72+52 = 148$.

Recall. The surface area of an $a\times b\times c$ box is $2(ab+bc+ca)$.

$2(4\cdot 5+5\cdot 6+6\cdot 4) = 2(20+30+24) = 148$. The two results agree because each 1×1 square on the outside of the box is exactly one painted small face of exactly one cube – we are simply counting the same collection in two ways.

c)

Step 6. Write the equation
For an $n\times n\times n$ cube: each face has $(n-2)^2$ middle squares, so the 6 faces give $6(n-2)^2$ cubes with exactly 1 painted face; the inside is a cube of $(n-2)^3$ small cubes. We need:

$$(n-2)^3 = 6(n-2)^2.$$

Step 7. Solve
Since $n\geq 3$, $(n-2)^2 > 0$; dividing both sides by $(n-2)^2$: $n-2=6$, i.e. $n=8$.

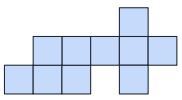
Common mistake. The condition $n\geq 3$ matters: for $n=2$ both numbers are 0 ("equal" in a trivial way), and we must not divide by 0.

Check: $n=8$: $6\cdot 6^2 = 216$ and $6^3 = 216$. Correct.

Answer: a) 8; 36; 52; 24 cubes. b) 148. c) $n=8$.

Problem 50 ★★

(Extension of Problems 19 and 25.) Ten unit squares (side 1) are joined to form a flat shape such that: adjacent squares share a whole side, and the shape is "in one piece" (from any square one can reach any other square through shared sides). The perimeter of the shape is the total length of the square sides that are not covered by another square.



an example

a) Show that the perimeter is always an even number.

b) What is the greatest possible perimeter?

c) What is the smallest possible perimeter?

d) Which numbers can be the perimeter of such a shape?

Detailed solution

Understanding the problem. Let k be the number of *shared sides* (sides where two squares are joined). Key idea: each shared side removes two square sides from the perimeter.

a)

Step 1. A formula for the perimeter

10 squares have $10 \cdot 4 = 40$ sides in total. Each shared side is a side of two squares and is not on the perimeter; every other square side is on the perimeter. Therefore

$$P = 40 - 2k.$$

40 and $2k$ are both even, so P is even.

b)

Step 2. A shape in one piece has at least 9 shared sides

Why is it true? Rebuild the shape by placing one square at a time: start with any square, and each time add a square next to one already placed (this is always possible because the shape is in one piece). Each added square creates at least 1 new shared side. There are 9 additions, so $k \geq 9$. □

Step 3. Conclusion

$P = 40 - 2k \leq 40 - 18 = 22$. Equality holds for a straight row 1×10: perimeter $2(10 + 1) = 22$. So the greatest perimeter is 22.

c)

Step 4. Compare with the enclosing rectangle

Let the smallest rectangle (with sides along the grid) containing the whole shape be $a \times b$ (a columns, b rows).

Why is it true? Every column of the enclosing rectangle contains at least one square. The highest square in the column has its top side on the perimeter, and the lowest square has its bottom side on the perimeter: each column contributes at least 2 units of horizontal sides. Likewise each row contributes at least 2 units of vertical sides. So $P \geq 2a + 2b = 2(a + b)$.□

Step 5. Bound $a + b$

The enclosing rectangle contains 10 squares, so $ab \geq 10$. If $a + b \leq 6$ then $ab \leq 9$ (pairs with sum 6: $1 \cdot 5 = 5$, $2 \cdot 4 = 8$, $3 \cdot 3 = 9$; smaller sums give even smaller products) – impossible. So $a + b \geq 7$ and $P \geq 14$.

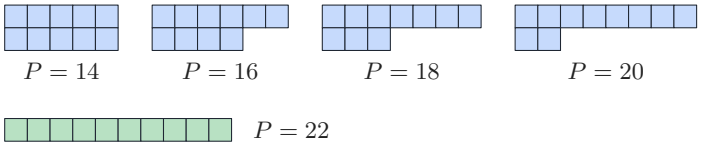
Step 6. A shape with perimeter 14 exists

The 2×5 rectangle: perimeter $2(2 + 5) = 14$. So the smallest perimeter is 14.

d)

Step 7. The possible values

From a), b), c): P is even and $14 \leq P \leq 22$, so $P \in \{14, 16, 18, 20, 22\}$. Each value is achieved by the following "two-row" shapes (using the idea of Problem 25: the boundary is a staircase, so the perimeter equals that of the enclosing rectangle):



Top row $5, 6, 7, 8$ squares, bottom row $5, 4, 3, 2$ squares: the enclosing rectangles are 5×2, 6×2, 7×2, 8×2, with perimeters $14, 16, 18, 20$; and the straight row gives 22.

Common mistake. "Compact" does not automatically mean a smaller perimeter: a 3×3 square with one extra square sticking out has perimeter $12 + 2 = 14$, the same as the 2×5 rectangle, not less.

Check: The $6 + 4$ shape: $k = 5 + 3 + 4 = 12$, $P = 40 - 24 = 16$. The $8 + 2$ shape: $k = 7 + 1 + 2 = 10$, $P = 40 - 20 = 20$. Correct.

Answer: a) $P = 40 - 2k$ **is even. b)** 22. **c)** 14. **d)** 14, 16, 18, 20, 22.